

INDUSTRY CERTIFICATION REPORT

Validation of Benchmark Artifacts for PhD Research Study



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Enhancing Generative AI Outcomes in Software Development Life Cycle:
A Unified Framework for Verifying and Improving LLM-Generated Specifications,
Designs, and Code

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1. Executive Summary

This Industry Certification Report serves as a formal validation document certifying that the benchmark software engineering artifacts (hereinafter referred to as the “Expected Outcomes” or “Ground-Truth Answer Key”) have been independently developed, professionally reviewed, and industrially validated by a team of senior software engineering practitioners from ProgosTech Pvt. Ltd. These benchmark artifacts are intended to serve as the reference standard (ground truth) for evaluating the outputs generated by the VeriGen framework, an automated tool developed as part of the above-mentioned PhD research study.

The VeriGen framework utilizes Large Language Models (LLMs) to automatically generate software engineering artifacts, including structured natural language specifications (in RUPPs template format), UML class diagrams, and UML sequence diagrams, from natural language problem descriptions. In any empirical evaluation of such a generative system, it is essential to have a validated, human-approved answer key against which the machine-generated outputs can be compared for correctness, completeness, consistency, and structural fidelity. The absence of such an answer key would render any quantitative or qualitative validation scientifically questionable.

Recognizing this critical requirement, the Department of Computer & Software Engineering at CEME, NUST, through its PhD supervisor Dr. Wasi Haider Butt, entered into a collaborative arrangement with ProgosTech Pvt. Ltd., a reputable software development company with proven expertise in enterprise-grade system analysis, design, and documentation. Under the technical leadership of Mr. Malik Hummayun (Head of Department - Web and Software Systems), a dedicated validation team comprising one Team Lead and two Senior Software Engineers was constituted. This team was tasked with independently creating and reviewing the Expected Outcomes for fifteen diverse software case studies.

This report documents the composition of the validation team, the rigorous methodology employed to ensure artifact quality, the independence of the benchmark creation process from the VeriGen tool, and the formal certification statement issued by the industry partner. Expected outcomes for fifteen case studies are available in the annexure attached under this document. By signing this report, both the academic supervisor and the industry lead affirm that the attached Expected Outcomes are fit for purpose as a validated answer key in the PhD research study.

2. Introduction and Background

The rapid advancement of generative artificial intelligence, particularly Large Language Models, has opened new frontiers in automating various phases of the software development life cycle. However, a fundamental challenge persists: how can researchers and practitioners objectively verify the quality, correctness, and utility of LLM-generated software artifacts such as requirements specifications, design models, and code? The PhD research undertaken by Mr. Mudassar Adeel Ahmed addresses this gap by proposing a unified framework (VeriGen) that not only generates such artifacts but also verifies and improves them through iterative feedback loops.

A critical component of any empirical validation of VeriGen is the existence of a reliable ground truth, a set of manually created, expertly reviewed “answer keys” that represent the correct or expected outcomes for a given set of input problem descriptions. For a controlled experiment, the researcher provides the same natural language problem description to both the VeriGen tool and a team of human experts. The outputs of VeriGen are then compared against the expert created answer key using appropriate metrics (e.g.,

syntactic similarity, structural overlap, semantic correctness). The validity of the entire experiment hinges on the quality and authority of this answer key.

Given the complexity of software engineering artifacts (which involve nuanced relationships, correct UML syntax, consistent naming, and traceability between textual descriptions and diagrams), it was determined that the answer key should be developed not by the researcher alone (to avoid bias) nor by academic peers alone (to ensure industrial realism), but by a team of experienced software practitioners from an active industry environment. ProgosTech Pvt. Ltd. was selected for this purpose due to its demonstrated track record in delivering complex, multi-platform software systems (e.g., e-commerce platforms, healthcare systems, real estate CRMs) and the direct availability of senior technical staff with expertise in both textual specifications and UML modeling.

This section establishes the context for the certification that follows. The benchmark artifacts cover fifteen distinct software systems, ranging from transportation booking systems (Zoom Car, Online Bus, Railway Reservation) to smart home systems (Digital Home), library systems (LMS, LIS), e-commerce platforms (EStore, Online Book Store, Online Pet Store), educational systems (College Registration, Puget, Online Discussion Group), and specialized systems (CCTNS, iCoot, Monitoring Operating System). The diversity of domains ensures that the answer key is not biased toward any single application type, thereby strengthening the generalizability of the VeriGen evaluation.

3. Scope of Benchmark Artifacts (Expected Outcomes)

The benchmark artifacts certified in this report are specifically the “Expected Outcomes” for the fifteen case studies. For each of the fifteen systems, the validation team produced or reviewed three interrelated artifacts:

Structured Natural Language Description (RUPPs-aligned): A detailed, step-by-step narrative describing the functional behavior of the system from the perspective of end-users (customers, members, administrators, etc.). These descriptions follow a pseudo-structured template resembling RUPPs, ensuring that they are both human-readable and machine-analyzable.

UML Class Diagram: A static structural model representing the key domain entities, their attributes, methods, and relationships (association, inheritance, aggregation, composition). Each class diagram is expressed using standard UML 2.0 notation and is intended to be complete enough to support code generation but concise enough to serve as a design-level specification.

UML Sequence Diagram: A dynamic interaction model showing the sequence of messages exchanged between objects (actors, system components) over time. The sequence diagrams capture the flow of events described in the textual narrative, including alternative paths and loops where appropriate.

The complete set of these artifacts is appended to this report in given Appendix. The artifacts are presented in the same order as the original document: Zoom Car Booking, Digital Home System, Library Management System, Monitoring Operating System, CCTNS, College Registration System, EStore, iCoot System, LIS, Online Bus Reservation, Online Book Store, Online Pet Store, Puget, Railway Reservation, and Online Discussion Group.

No modifications have been made to the original artifacts except for formatting consistency. The validation team confirmed that each artifact is internally consistent (e.g., a class named in the textual description

appears in the class diagram, and the sequence diagram refers to instances of those classes) and externally complete (i.e., no major functionality described in the text is missing from the diagrams).

4. Industry Validation Team

The validation of the benchmark artifacts was carried out under the authority and supervision of ProgosTech Pvt. Ltd. The team comprised the following experienced professionals, each bringing a specific skill set essential to the review process.

4.1. Project Lead and Industry Certifier

Mr. Malik Hummayun

Head of Department - Web and Software Systems, ProgosTech Pvt. Ltd.

Mr. Hummayun is a results-driven Technical Project Manager with over 13 years of experience in software development and team leadership. He has led end-to-end development of mobile and web applications for international clients, including Tammana (Dubai), Aliaaj (Pakistan), and Elabelz (UAE). His technical expertise spans iOS (Swift, Objective-C), Android (Java), React Native, and Agile methodologies. For this validation effort, Mr. Hummayun defined the overall quality criteria for the answer key, chaired the review meetings, and resolved any discrepancies between team members. He also verified that the benchmark artifacts align with real-world software documentation standards used in the industry. His full curriculum vitae is provided in Appendix B for reference and contact purposes.

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4.2. Team Lead Architecture and Consistency

Mr. Bilal Ahmed

Lead Software Architect, ProgosTech Pvt. Ltd.

Mr. Ahmed has over 10 years of experience in designing scalable software architectures for enterprise applications. In this validation, he was responsible for maintaining consistency across all fifteen case studies. He ensured that UML notation was used correctly, that class diagrams followed a uniform layout and naming convention, and that the complexity of each model was appropriate for the corresponding textual description. He also cross-checked that sequence diagrams properly reflected the interactions implied by the class diagrams and use-case narratives.

4.3. Senior Software Engineer – System Analysis (Textual Specifications)

Ms. Fatima Zafar

Senior Software Engineer – System Analysis, ProgosTech Pvt. Ltd.

Ms. Zafar specializes in requirements engineering and natural language processing for software specifications. She was responsible for authoring the structured natural language descriptions (in RUPPs-like format) for seven of the fifteen case studies and reviewing the remaining eight. Her work ensured that each description included step-by-step interactions, error handling, and references to help facilities. She also verified traceability between the textual narrative and the corresponding UML diagrams.

4.4. Senior Software Engineer – UML Modeling

Mr. Usman Chaudhary

Senior Software Engineer – Modeling, ProgosTech Pvt. Ltd.

Mr. Chaudhary has extensive experience in creating UML diagrams for real-time and business systems using enterprise modeling tools (e.g., Lucidchart, Draw.io, Enterprise Architect). He created the class diagrams and sequence diagrams for eight of the fifteen case studies and reviewed the remaining seven. He paid particular attention to correct element of concerns, accurate representation of inheritance and composition, proper use of activation bars and return messages in sequence diagrams, and overall aesthetic clarity suitable for academic publication.

All team members declared that they had no conflict of interest with the PhD research or the VeriGen tool. Their work was conducted independently, without access to any VeriGen-generated outputs, ensuring that the benchmark artifacts truly represent a human-created gold standard.

5. Validation Methodology

The validation team followed a structured, multi-phase methodology to ensure the highest possible quality of the benchmark artifacts. Each phase involved specific activities, deliverables, and quality gates.

5.1. Phase 1: Independent Artifact Creation or Review

For each of the fifteen case studies, the team first determined whether the existing artifacts (as provided in the original document) were complete and correct, or whether they required revision. The original document contained a mix of well-formed and partially-formed descriptions. The team's approach was as follows:

For case studies where the textual description was detailed and the diagrams were present, the team performed a thorough review and made minor corrections (e.g., fixing UML syntax, clarifying ambiguous steps).

For case studies where the textual description was present but diagrams were missing or incomplete, the senior modeler (Mr. Chaudhary) created new diagrams from scratch based solely on the textual description.

For case studies where the textual description itself needed improvement, the system analyst (Ms. Zafar) revised the text to meet RUPPs standards.

All original content was retained as much as possible; only necessary corrections were made to eliminate ambiguity or syntax errors.

5.2. Phase 2: Consistency and Traceability Review

The Team Lead (Mr. Ahmed) conducted a systematic cross-check across all artifacts. The following consistency criteria were applied:

Every actor named in the textual description (e.g., Customer, Admin, Librarian) must appear as an actor in the sequence diagrams and potentially as a class boundary.

Every major noun that represents persistent data (e.g., Car, Booking, Payment) must appear as a class in the class diagram.

Every key use case described in the text must have a corresponding sequence diagram or at least a fragment thereof.

Class diagrams must not contain attributes or methods that are never referenced in the textual description or sequence diagrams (no “orphaned” elements). Sequence diagrams must not refer to classes or methods that do not exist in the class diagram. Any violation of these criteria was logged and corrected by the original author or the team lead.

5.3. Phase 3: UML Compliance and Notation Verification

All UML diagrams were verified against the official UML 2.0 specification for correct notation. Specific checks included:

Class diagrams: correct diamond notation for composition (filled diamond) vs. aggregation (empty diamond); correct arrowhead for inheritance (hollow triangle); etc.

Sequence diagrams: correct actor lifeline notation; correct use of alt, opt, loop fragments; synchronous vs. asynchronous message arrows; return messages shown with dashed arrows where appropriate.

Diagrams that failed this verification were redrawn using a standard UML tool to ensure publication-ready quality.

5.4. Phase 4: Industrial Realism Assessment

The Project Lead (Mr. Hummayun) reviewed the artifacts from an industry practitioner’s perspective. He assessed whether the expected outcomes reflected the level of detail and realism that would be expected in a real software requirements specification or design document produced by a competent development team. For example, he checked that error conditions (e.g., invalid login, out-of-stock items, cancellation fines) were adequately captured, not just the “happy path.” He also ensured that the artifacts were neither overly simplistic (which would render validation trivial) nor overly complex (which would make them impossible to replicate by an LLM).

5.5. Phase 5: Final Approval and Sign-off

After all corrections were made and all team members had approved the revised artifacts, a final walkthrough was conducted with the PhD supervisor (Dr. Wasi Haider Butt) to confirm that the artifacts met academic expectations. The supervisor provided feedback on clarity and completeness, which was incorporated. The final approved set of artifacts was then compiled into Appendix, and this certification report was prepared for signature.

6. Independence Statement

To maintain the scientific integrity of the PhD research, it was imperative that the benchmark artifacts (answer key) be developed completely independently of the VeriGen tool and without any influence from the tool’s outputs. The validation team hereby certifies the following:

No VeriGen-generated content was used as a source or reference during the creation or revision of any of the fifteen Expected Outcomes. The team relied exclusively on the original natural language descriptions provided in this document and their own professional judgment.

The VeriGen tool was not available to the validation team at the time of artifact creation or review. The team had no knowledge of the internal algorithms, training data, or generated outputs of VeriGen.

No bidirectional feedback occurred between the validation team and the PhD candidate regarding the content of the benchmark artifacts while the artifacts were being finalized. The candidate only provided the raw source material (the original case study descriptions) and received the final answer key after certification.

Any corrections made to the original artifacts were based solely on standard software engineering principles, UML specification rules, and the team's industrial experience, not on a desire to make the answer key easier or harder for VeriGen to match.

This independence guarantees that the subsequent evaluation of VeriGen against these benchmark artifacts will be a fair, unbiased, and scientifically valid comparison between machine-generated and human-created outcomes.

7. Certification Statement

Based on the validation methodology described above and the independence assurance provided, the undersigned representatives of ProgosTech Pvt. Ltd. and the National University of Sciences and Technology (NUST) hereby certify the following:

The Expected Outcomes (structured natural language descriptions, UML class diagrams, and UML sequence diagrams) for the fifteen case studies listed in Appendix constitute a complete, correct, and professionally reviewed set of software engineering artifacts.

These artifacts are suitable for use as benchmark answer keys (ground truth) in the controlled experimental evaluation of the VeriGen framework, as part of the PhD research study titled "Enhancing Generative AI Outcomes in Software Development Life Cycle: A Unified Framework for Verifying and Improving LLM-Generated Specifications, Designs, and Code" by Mr. Mudassar Adeel Ahmed.

The artifacts reflect realistic industrial standards of documentation and design, having been created and validated by senior software practitioners with extensive experience in commercial software development.

The artifacts were developed independently of the VeriGen tool, ensuring no bias in the validation process.

Permission is granted to the PhD candidate, NUST, and any thesis examination committee to reproduce, cite, and append these artifacts.

This certification remains valid for the duration of the PhD examination process and any subsequent revisions or resubmissions of the thesis, unless explicitly revoked in writing by both signatories.

8. Contact Information for Verification

For any inquiries regarding this certification, the identity of the validation team, or the authenticity of the appended artifacts, please contact:

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9. Appendix

Expected Outcomes of fifteen Case Studies (including structured natural language descriptions, UML class diagrams, and UML sequence diagrams for each).

Approval and Certification



Dr. Wasi Haider Butt
Professor
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Rawalpindi, Pakistan

Dr. Wasi Haider Butt
Professor
Department of Computer & Software Engineering
College of E&ME, NUST



Malik Muhammad Hummayun
Head of Department
Web, Mobile Apps & Software Systems
ProgosTech Pvt. Ltd

Library Management System Description

The Member clicks the log-in button on the Home Page. The system displays the log-in page. The member enters his login Id with password. The member clicks on the confirm button. The system checks that all of the required information was entered. If the entered information is wrong then system asks member to reenter the details. The system validates the entered information against the tables stored in the database. Member must be logged in to the system. The member clicks on View user details. The system opens a page showing the details of the member. The details include the total number of issued books, date of issue, return date, fine to be paid. The member closes the page. The system shows OK message. The books should be stored in database. The books must be ready to retrieve. The system asks user to login. The system identifies the type of user- member, guest or administrator. The system shows the categories to browse. The user selects a category of books to view. If no category is selected by the user then the system again asks user to select a category. The system checks the books in the database. The system retrieves all the books falling in that category. The user selects the desired books. The system shows the details of the selected books. The system asks user to print the details. If User does not want to print the details then user can ignore the step. User can reserve a book by inputting the relevant details & the librarian can also reserve a book for a member. User should be logged into the system. User should have correct book Id. Books should be available to reserve. The system asks user to login. The system identifies the type of user- member, guest or administrator. The system shows the categories to browse. The user selects book to reserve. If no book is selected by the user then the system again asks user to select a book to reserve. The user enters Book Id to reserve. If the book Id is wrong then system asks user to recheck the book id. The system checks the books in the database. If the selected book is already reserved on another Id then system asks user to select another book. Members should be logged into the system. Guest user can also search books. Book should be available to search. The system shows the categories to browse. The member selects a category of searching a book. If no category is selected by the member then the system again asks user to select a category. The system checks the books in the database. The system retrieves all the books falling in that category. Member should give the member Id to the librarian. Books should be available to issue. The librarian Checks the availability of the books. The librarian also checks total number of books issued on that Id. Librarian issues the book. The librarian cannot issue the books if the user has three books issued on his Id. The system updates the information in database. Librarian should be logged into the system. Member should have borrowed books. Member should give the member Id to the librarian. The member returns the book. The librarian enters book id , member id in the system. If the entered book id is incorrect then system asks to re-enter the book id. The system prompts a message that the book with book id is successfully returned. Members should be stored in the database. Members should be available to retrieve by the system. Member must be logged in to the system. Guest can also view members. The user clicks on View members. The system opens a page showing the details of the member. The details include name of member, the total number of issued books, date of issue, return date, fine to be paid. The member closes the page. Librarian should be logged into the system. Books should be available to remove. Book details should be available to add or remove books in the database. The librarian has option of adding or removing a book in database. The system asks librarian to add or remove the book. The librarian adds a book. The system asks librarian to enter all the required details about the new book to be added. The librarian enters the details. If the librarian selects to remove a book then the book should be outdated. The system administrator that all the information has

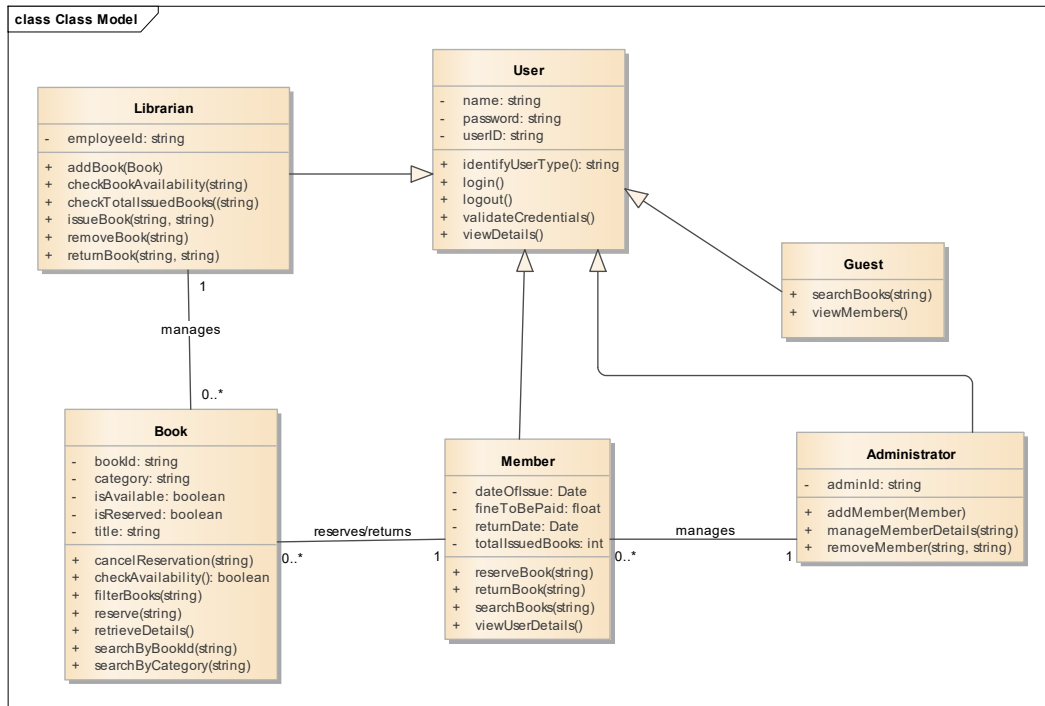
been correctly provided. Administrator should be logged into the system. Member should be available to remove. Member details should be available to add or remove member in the database. The system asks administrator to add or remove a member. The administrator selects to add a member. The system asks administrator to enter all the required details about the new member to be added. The administrator enters the details. If the administrator selects to remove a member then a valid reason of removal is required. The system validates that all the information has been correctly provided.

Library Management System (SNL)

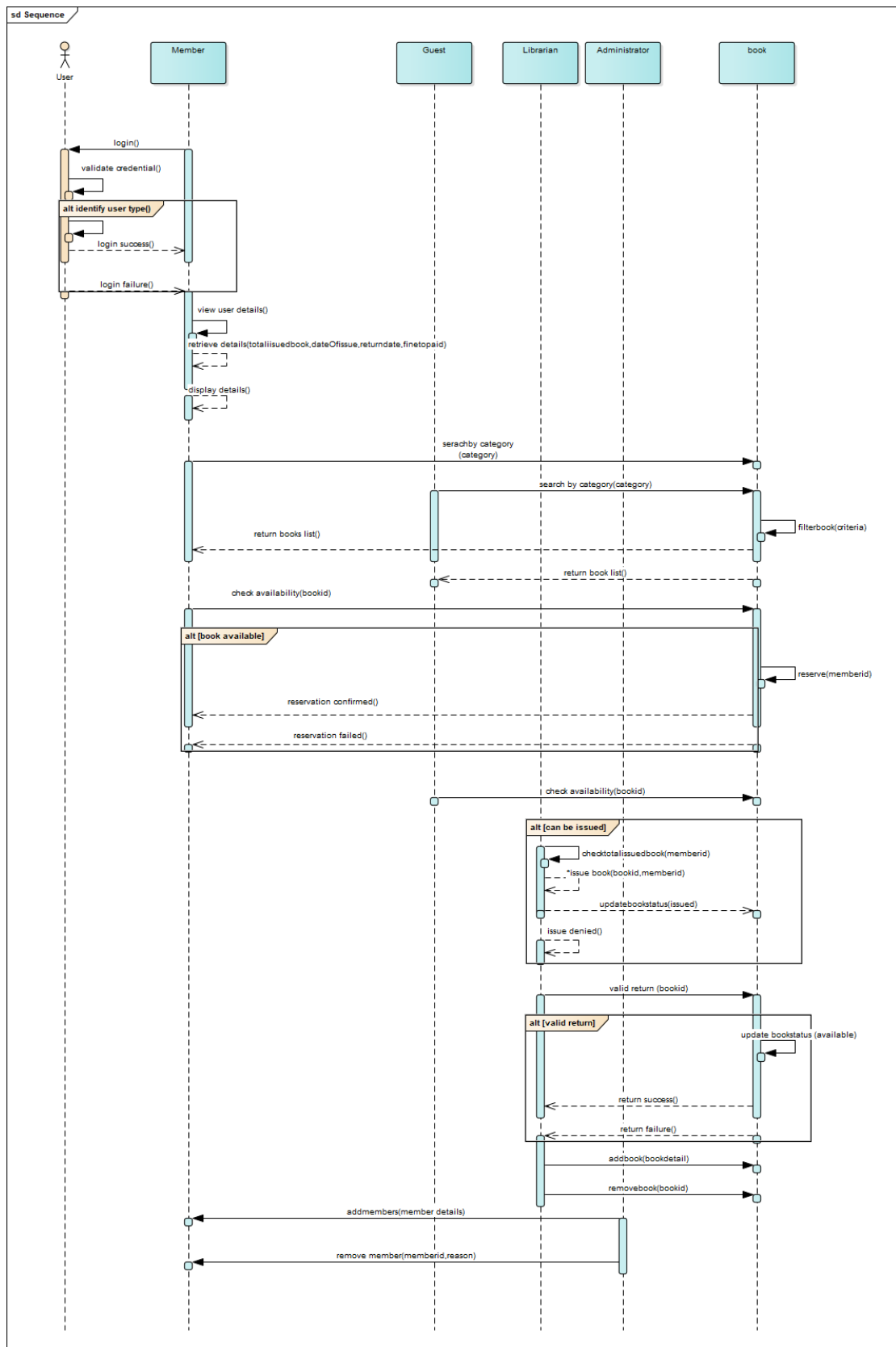
The system shall provide member with the ability to click the login button on the home page. The system shall be able to display the login page. The system shall provide member with the ability to enter his login ID with password. The system shall provide member with the ability to click on the confirm button. The system shall be able to check that all of the required information entered. If the entered information is wrong then the system shall be able to ask member to reenter details. The system shall be able to validate the entered information against the tables stored in the database. The system shall provide member with the ability to logged in. The system shall provide member with the ability to click on view user details. The system shall be able to open a page showing the details of the member. The system shall display the details including the total number of issued books, date of issue, return date, and fine to be paid. The system shall provide member with the ability to close the page. The system shall be able to show OK message. The system shall be able to stored books in database. The system shall be able to retrieve ready books. The system shall be able to ask user login. The system shall be able to identify the type of user member, guest or administrator. The system shall be able to show the categories to browse. The system shall provide user with the ability to select a category of books to view. If no category is selected by the user then the system shall be able to ask again user to select category. The system shall be able to check the books in the database. The system shall be able to retrieve all the books falling in that category. The system shall provide user with the ability to select the desired books. The system shall be able to show the details of the selected books. The system shall be able to ask user to print the details. If user does not want to print the details then the system shall provide user with the ability to ignore the step. The system shall provide user with the ability to reserve a book by inputting the relevant details. The system shall provide librarian with the ability to also reserve a book for a member. The system shall provide user with the ability to logged in. The system shall provide user with the ability to have correct book ID. The system shall be able to reserve available books. The system shall be able to ask user login. The system shall be able to identify the type of user member, guest or administrator. The system shall be able to show the categories to browse. The system shall provide user with the ability to select book to reserve. If no book is selected by the user then the system shall be able to ask again user to select book to reserve. The system shall provide user with the ability to enter book ID to reserve. If the book ID is wrong then the system shall be able to ask user to recheck book ID. The system shall be able to check books in database. If the selected book is already reserved on another ID then the system shall be able to ask user to select book. The system shall provide member with the ability to logged in. The system shall provide guest_user with the ability to also search books. The system shall be able to search available book. The system shall be able to show the categories to browse. The system shall provide member with the

ability to select a category of searching a book. If no category is selected by the member then the system shall be able to ask again user to select category. The system shall be able to check the books in the database. The system shall be able to retrieve all the books falling in that category. The system shall provide member with the ability to give the member ID to the librarian. The system shall be able to issue available books. The system shall provide librarian with the ability to check the availability of the books. The system shall provide librarian with the ability to check total number of books issued on that ID. The system shall provide user with the ability to not issue the books if the has three books issued on his ID. The system shall provide librarian with the ability to issue the book. The system shall update the information in database. The system shall provide librarian with the ability to logged in. The system shall provide member with the ability to borrow books. The system shall provide member with the ability to give the member ID to the librarian. The system shall provide member with the ability to return the book. The system shall provide librarian with the ability to enter book ID, member ID in the system. If the entered book ID is incorrect then the system shall be able to be asks to reenter book ID. The system shall be able to prompt a message that the book with book ID has been successfully returned. The system shall be able to stored members in database. The system shall be able to retrieve available members. The system shall provide member with the ability to logged in. The system shall provide guest with the ability to also view members. The system shall provide user with the ability to click on view members. The system shall be able to open a page showing the details of the member. The system shall display the details including name of member, the total number of issued books, date of issue, return date, and fine to be paid. The system shall provide member with the ability to close the page. The system shall provide librarian with the ability to logged in. The system shall be able to remove available books. The system shall be able to add or remove available book details in the database. The system shall provide librarian with the ability to have option of adding or removing a book in database. The system shall be able to ask librarian to add or remove the book. The system shall be able to ask librarian to enter all the required details about the new book to added. The system shall provide librarian with the ability to add a book. The system shall provide librarian with the ability to enter the details. If the librarian selects to remove a book then the system shall be able to remove the outdated book. The system shall provide system_administrator with the ability to correct all the information has been provided. The system shall provide administrator with the ability to logged in. The system shall provide administrator with the ability to add or remove member details in the database. The system shall be able to ask administrator to add or remove a member. The system shall provide administrator with the ability to select to add a member. The system shall be able to ask administrator to enter all the required details about the new member to added. The system shall provide administrator with the ability to enter the details. If the administrator selects to remove a member then the system shall be able to remove valid reason of removal required. The system shall be able to validate that all the information correctly provided.

Library Management System Class Diagram



Library Management System Sequence Diagram



Monitoring Operating System Description

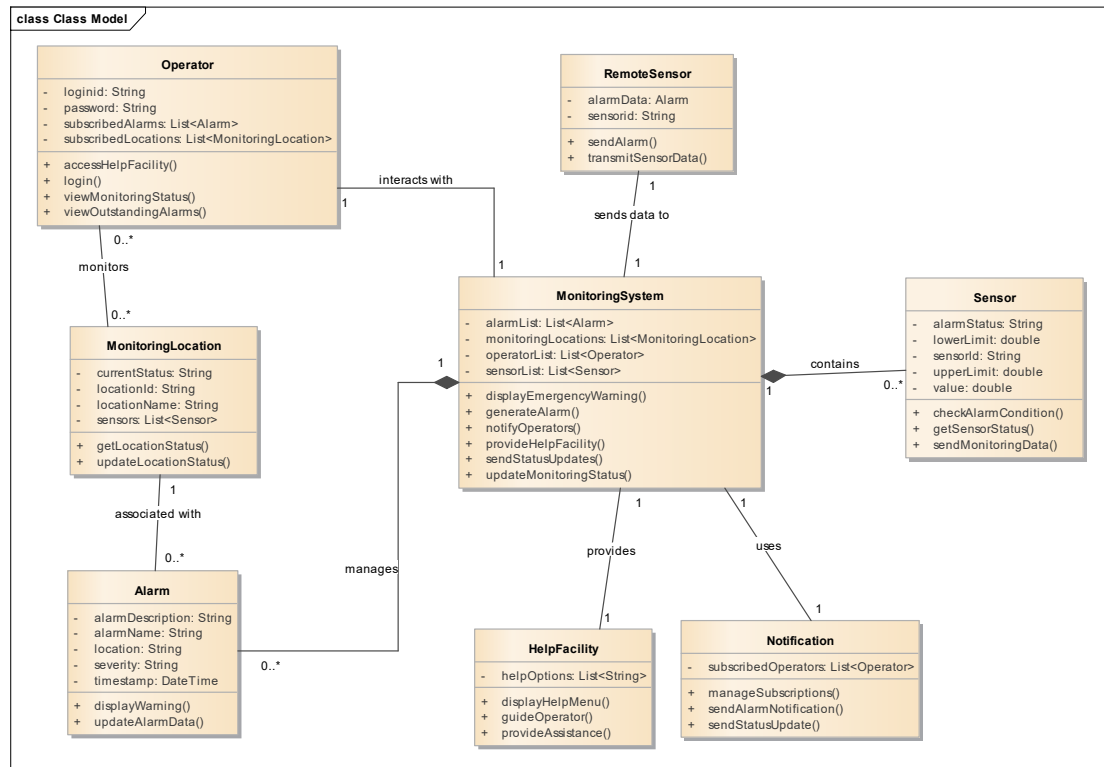
The Operator is already logged in. The Operator requests to view the status of a monitoring location. If an Emergency situation occurs then System displays emergency warning message to operator. The system displays the monitoring status as follows: Sensor status for each sensor (value, upper limit, lower limit, alarm status). Monitoring status has been displayed. The Operator is logged in before starting the process. The Operator requests to view the outstanding alarms. If an Emergency situation occurs then System displays emergency warning message to operator. The system displays the outstanding alarms. For each alarm, the system displays the name of the alarm, alarm description, location of alarm, severity of alarm (high, medium, low). Outstanding alarms have been displayed. The remote system is operational in advance. The sensors send new monitoring data to the system. If an Emergency situation occurs then System displays emergency warning message to operator. The system updates the monitoring status as follows: Sensor status for each sensor (value, upper limit, lower limit, alarm status). The system sends new monitoring status to Monitoring-Operators who have subscribed to receive status updates. Monitoring status has been updated. If an alarm condition is detected then an alarm is generated. Operators are notified of new alarms to which they have subscribed. If Emergency situation is detected then System displays emergency warning message to operator. The external sensor is operational already. The Remote Sensor sends an alarm to the system. If the alarm is severe then a display with flashing warning is done. The system updates the alarm data. The system stores the name of the alarm, alarm description, location of alarm, severity of alarm (high, medium, low). Alarm data have been updated. If the Operator is stuck somewhere in the process then he has an option to use help facility provided by the system. The Monitoring-Operator chooses help facility option. Screen displays a message asking the type of help he wants. Depending on the selection, the system helps him in proceeding with the system.

Monitoring Operating System (SNL)

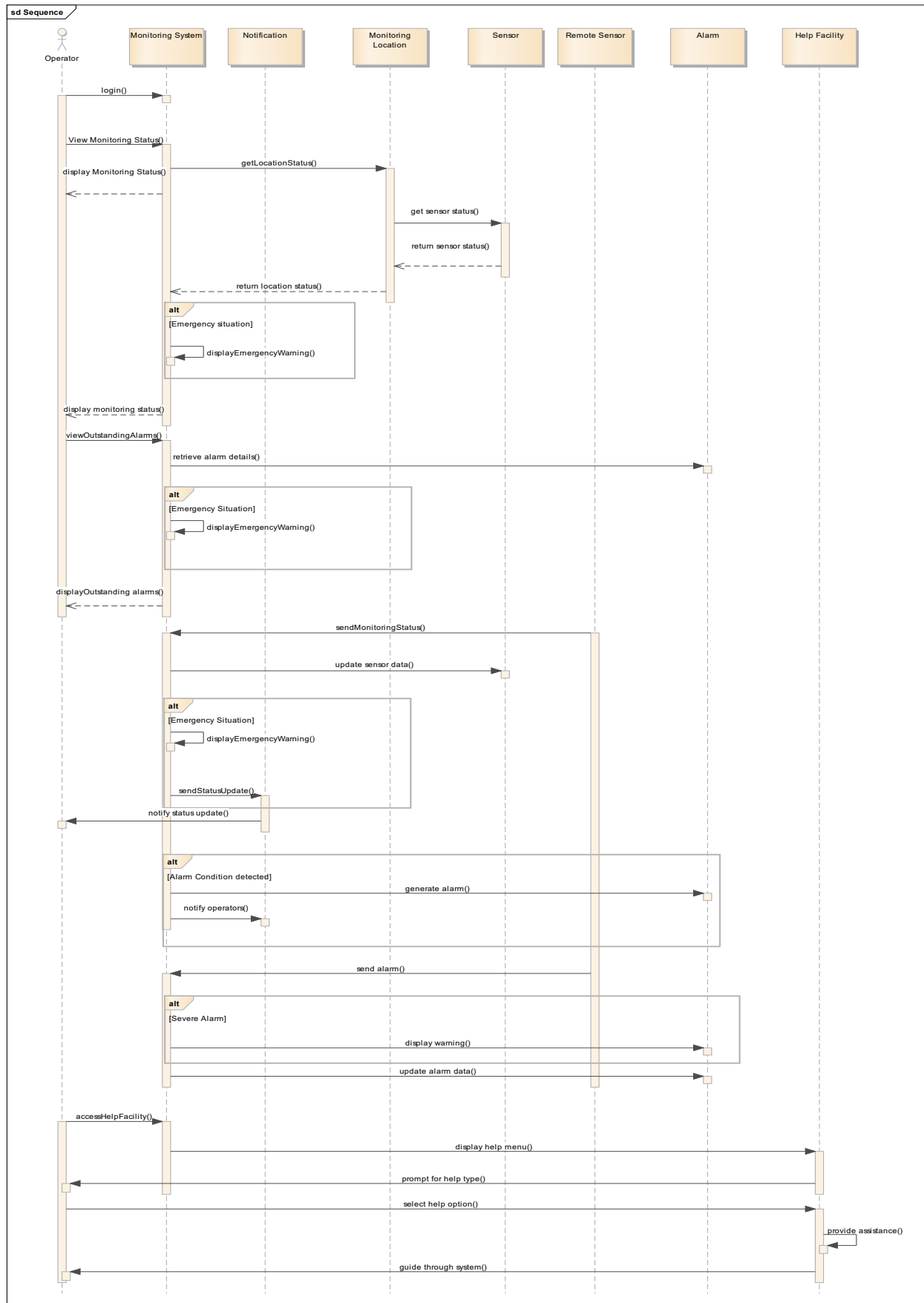
The system shall be able to ensure the operator is logged into the system. The system shall provide operator with the ability to request to view the status of a monitoring location. If an emergency situation occurs then the system shall be able to display an emergency warning message to the operator. The system shall be able to display the monitoring status as follows: sensor status for each sensor value, upper limit, lower limit, alarm status. Monitoring status has been displayed. The system shall provide operator with the ability to request to view the outstanding alarms. The system shall be able to display the outstanding alarms. For each alarm, the system shall be able to display the name of the alarm, alarm description, location of alarm, severity of alarm high, medium, low. Outstanding alarms have been displayed. The system shall be able to ensure the system is operational in advance. The sensors send new monitoring data to the system. The system shall be able to update the monitoring status including sensor value, upper limit, lower limit, and alarm status. The system shall be able to send new monitoring status to Monitoring-Operators who have subscribed to receive status updates. Monitoring status has been updated. If an alarm condition is detected then the system shall be able to generate an alarm. The system shall be able to notify operators of new alarms to which they have subscribed. If an emergency situation is detected then the system shall be able to display an emergency warning message to the operator. The system shall provide remote sensor with the ability to send an alarm to the system. If the alarm is severe then the system shall be able to display a flashing warning. The system shall be able to update alarm data. The system shall be able to store alarm details including name, description, location of alarm, severity (high, medium, low). Alarm data have been updated. If the operator is stuck somewhere in the process then the system shall provide operator with the ability to use help facility option provided by the system. The system shall

provide monitoring operator with the ability to choose help facility option. The system shall be able to display a message asking what type of help the monitoring operator wants. The system shall be able to help the operator in proceeding with the system.

Monitoring Operating System Class Diagram



Monitoring Operating System Sequence Diagram



Zoom Car Booking Description

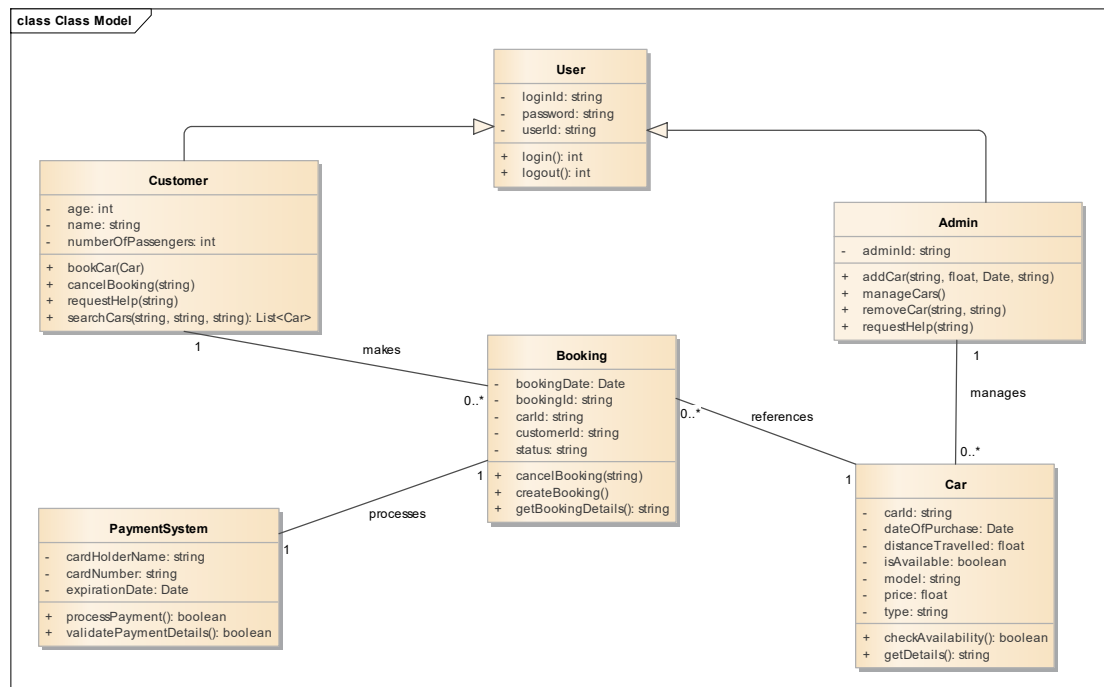
The system displays the log-in page. The customer enters his login Id with password. The customer clicks on the confirm button. The system checks that all of the required information was entered. If the entered information is wrong then system asks customer to reenter the details. The system validates the entered information. The Admin clicks the log-in button on the Home Page. The system displays the log-in page. The Admin enters his login Id with password. The admin clicks on the confirm button. The system checks that all of the required information was entered. If the entered information is wrong then system asks Admin to reenter the details. The system validates the entered information against the tables stored in the database. The customer already has navigated to the main options screen. The customer enters the details of source station, destination station to search cars on that route. The customer also enters the type of car- hatch back, sedan etc he wants to book. Customer enters all the information- name, age, number of passengers, source station, destination station. The customer can use help facility to know how to fill the information, HelpFacility usecase is carried out. He clicks the "Search" button. If all the required information is not entered by the customer then the system highlights the blank fields to be entered. After searching, the System displays a list of all the available cars that matches the customer's input, each car in the list has an associated price. Customer must have searched for the required car. System displays a screen showing the details of the booked car. The customer verifies the details. He clicks the "Submit" button. System displays a screen with input fields for: a card number, cardholder name, card expiration date. Customer enters the details for the card number, the cardholder name , the card expiration date and he clicks the "Submit" button. If customer is stuck in the process then he can use help facility to know how to fill the information, HelpFacility usecase is carried out. System displays the success message. A booking has already been made and customer must have successfully navigated to the main options screen. Customer Selects the "Cancel booking" option. System displays a screen with an input field for the reason of cancellation. He clicks the "Submit" button. The customer can use help facility, HelpFacility use case is carried out. System displays the details of the booking. Customer Selects the "Process Cancellation" option. System displays a message "Booking successfully cancelled" and then system displays the main options screen again. The customer clicks on help facility. The system displays a list of facilities. The customer clicks on the type of help required. The system shows the instructions. The admin manages the car. The system asks Admin whether he wants to Add a new car or Remove an outdated car from the system. The Admin selects to add car, AddCar usecase is carried out. If Admin selects to remove an outdated car, then RemoveCar usecase is carried out. The Admin already has navigated to the main options screen. The system validates the log in information. The admin selects on adding the car. The system asks admin to fill all the required information for adding a car in the database- car model, distance travelled so far, date of purchase, type of car etc. The admin fills the information. The system displays a success message on the screen. If all the required information is not filled by Admin then the system shows an error message. The Admin already has navigated to the main options screen. The system validates the log in information. The admin selects to remove car. The system asks admin to fill all the required information for removing a car in the database. The system asks admin to fill all the required information for removing a car in the database- the reason of removing that car, car model, distance travelled so far, date of purchase, type of car etc. The system displays a success message on the screen. If all the required information is not filled by Admin then the system shows an error message.

Zoom Car Booking (SNL)

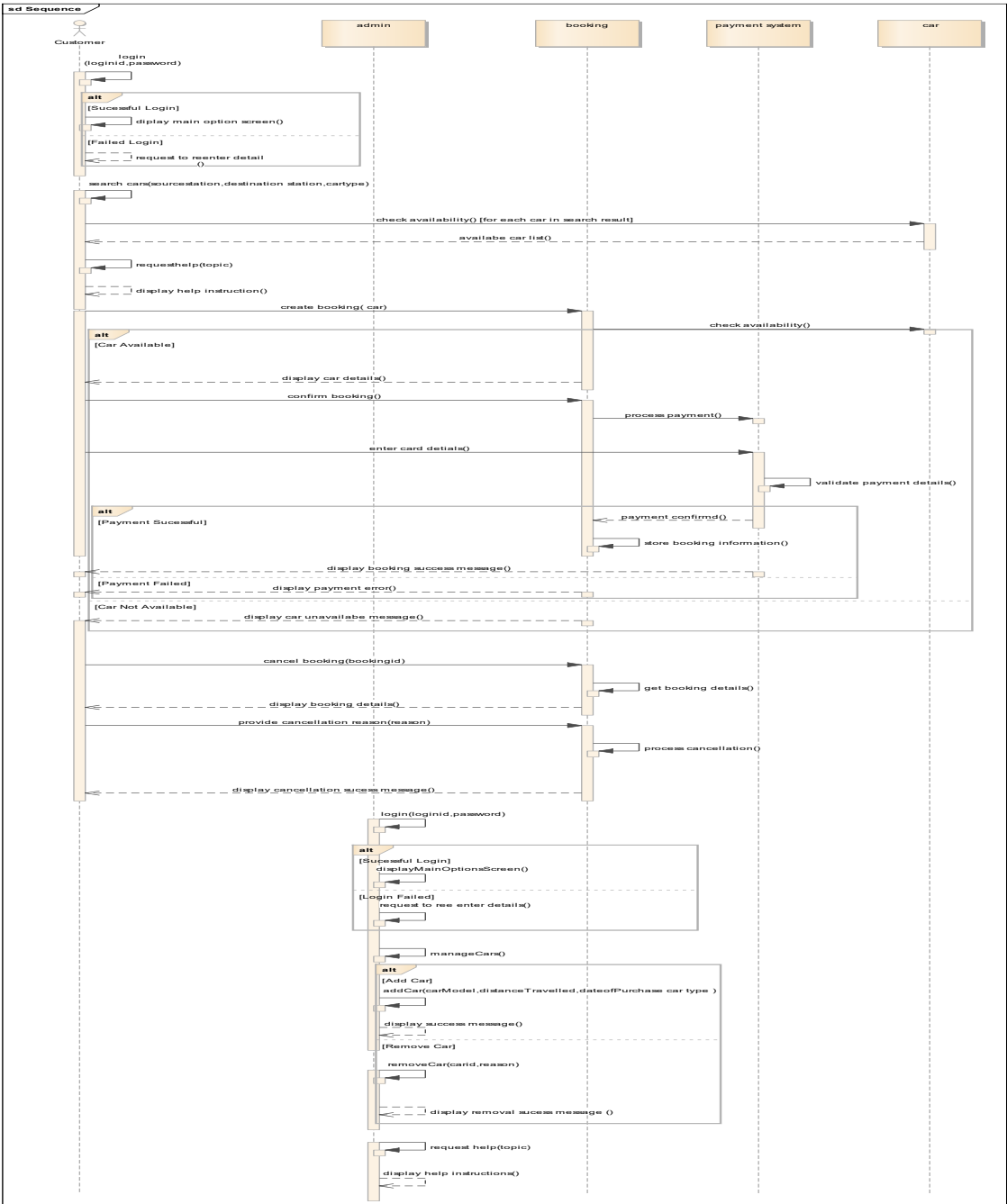
The system shall be able to display the login page. The system shall provide customer with the ability to enter his login ID with password. The system shall provide customer with the ability to click on the confirm button. The system shall be able to check that all of the required information entered. If the entered information is wrong then the system shall be able to ask customer to reenter details. The system shall be able to validate the entered information. The system shall provide admin with the ability to click the login button on the home page. The system shall be able to display the login page. The system shall provide admin with the ability to enter his login ID with password. The system shall provide admin with the ability to click on the confirm button. The system shall be able to check that all of the required information entered. If the entered information is wrong then the system shall be able to ask admin to reenter details. The system shall be able to validate the entered information against the tables stored in the database. The system shall provide customer with the ability to already navigate to the main options screen. The system shall provide customer with the ability to enter the details of source station, destination station to search cars on that route. The system shall provide customer with the ability to also enter the type of car hatch back, sedan etc admin wants to book. The system shall provide customer with the ability to enter all the information name, age, number of passengers, source station, destination station. The system shall provide customer with the ability to use help facility to know how to fill the information, HelpFacility usecase carried out. The system shall provide customer with the ability to click search button. If all the required information is not entered by the customer then the system shall be able to highlight the blank fields to be entered. The system shall be able to display a list of all the available cars that matches the customers input, each in the list has an associated price. The system shall be able to verify that the customer has searched for the required car. The system shall be able to display a screen showing the details of the booked car. The system shall provide customer with the ability to verify the details. The system shall provide customer with the ability to click submit button. The system shall be able to display a screen with input fields for a card number, cardholder name, card expiration date. The system shall provide customer with the ability to enter the details for the card number, the cardholder name, the card expiration date. The system shall provide customer with the ability to click submit button. If customer is stuck in the process then the system shall provide customer with the ability to use help facility to know how to fill the information, HelpFacility usecase carried out. The system shall be able to display the success message. The system shall provide customer with the ability to successfully navigate to the main options screen. The system shall provide customer with the ability to select the cancel booking option. The system shall be able to display a screen with an input field for the reason of cancellation. The system shall provide customer with the ability to click submit button. The system shall provide customer with the ability to use help facility, HelpFacility use case carried out. The system shall be able to display the details of the booking. The system shall provide customer with the ability to select the process cancellation option. The system shall be able to display a message booking successfully cancelled and then displays the main options screen again. The system shall provide customer with the ability to click on help facility. The system shall be able to display a list of facilities. The system shall provide customer with the ability to click on the type of help

required. The system shall be able to show the instructions. The system shall provide admin with the ability to manage the car. The system shall be able to ask admin whether he wants to add a new car or remove an outdated car from the system. The system shall provide admin with the ability to select to add car, AddCar usecase carried out. If admin selects to remove an outdated car, then the system shall be able to carry out the RemoveCar use case. The system shall provide admin with the ability to already navigate to the main options screen. The system shall be able to validate the log in information. The system shall provide admin with the ability to select on adding the car. The system shall be able to ask admin to fill all the required information for adding a car in the database car model, travelled so far, date of purchase, type of car etc. The system shall provide admin with the ability to fill the information. The system shall be able to display a success message on the screen. If all the required information is not filled by admin then the system shall be able to show an error message. The system shall provide admin with the ability to already navigate to the main options screen. The system shall be able to validate the log in information. The system shall provide admin with the ability to select to remove car. The system shall be able to ask admin to fill all the required information for removing a car in the database. The system shall be able to ask admin to fill all the required information for removing a car in the database the reason of removing that car, car model, travelled so far, date of purchase, type of car etc. The system shall be able to display a success message on the screen. If all the required information is not filled by admin then the system shall be able to show an error message.

Zoom Car Booking Class Diagram



Zoom Car Booking Sequence Diagram



Digital Home System Description

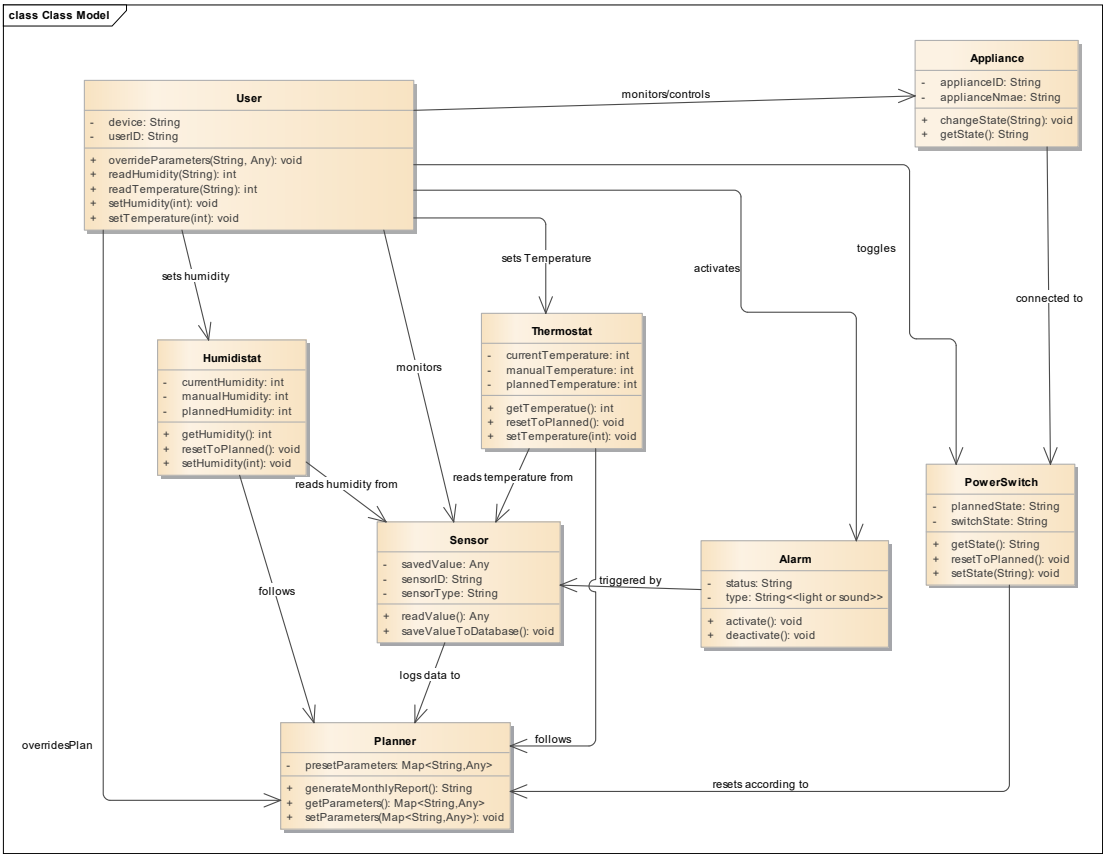
The System allows a web ready computer, cell phone or PDA to control a home's temperature, humidity, lights, security, and the state of small appliances. Users monitor and control home devices and systems. The system equips with various environmental controllers and sensors. The sensor reads and saved value in the home database. The user read the temperature at a thermostat position. The user set the thermostat temperatures to between 60 ° F and 80 ° F, inclusive, at one degree increments. If a thermostat device allows a user to make a manual temperature setting, the setting remains in effect until the end of the planned or default time period, at which time the planned or default setting will be used for the next time period. The system compatibles with a centralized HVAC (Heating, Ventilation and Air Conditioning). The system adheres to the standards, policies and procedures of the American Society of Heating, Refrigerating and Air-Conditioning Engineers [ASHRAE 2010]. The user monitor and control a home's humidity from any location, using a web ready computer, cell phone, or PDA. The user read the humidity at a humidistat position. The user sets the humidity level for a humidistat, from 30% to 60%, inclusive a 1% increments. The system uses wireless signals to communicate, through the master control unit, with the humidistats. The system consists of contact sensors and a set security alarms. The system manages up to fifty door and window contact sensors. The system activates both light and sound alarms one sound alarm and one light alarm subsystem, with multiple lights. The system provides information about the state of a power switch (OFF or ON), indicating the whether an appliance connected to the power switch is OFF or ON. The system changes the state of a power switch (OFF to ON, or ON to OFF), in turn changing the state of an appliance connected to the power switch. If a user changes the state of power switch device manually, the device remains in that state until the end of the planned or default time period, at which time the planned or default setting will be used for the next time period. The system set various preset home parameters (temperature, humidity, security contacts, and on/off appliance/light status) for certain time periods. System provides a monthly planner. The user overrides planned parameter values, through the DH website, or if available, through manual switches on household devices. The system provides a report on the management and control of the home.

Digital Home System (SNL)

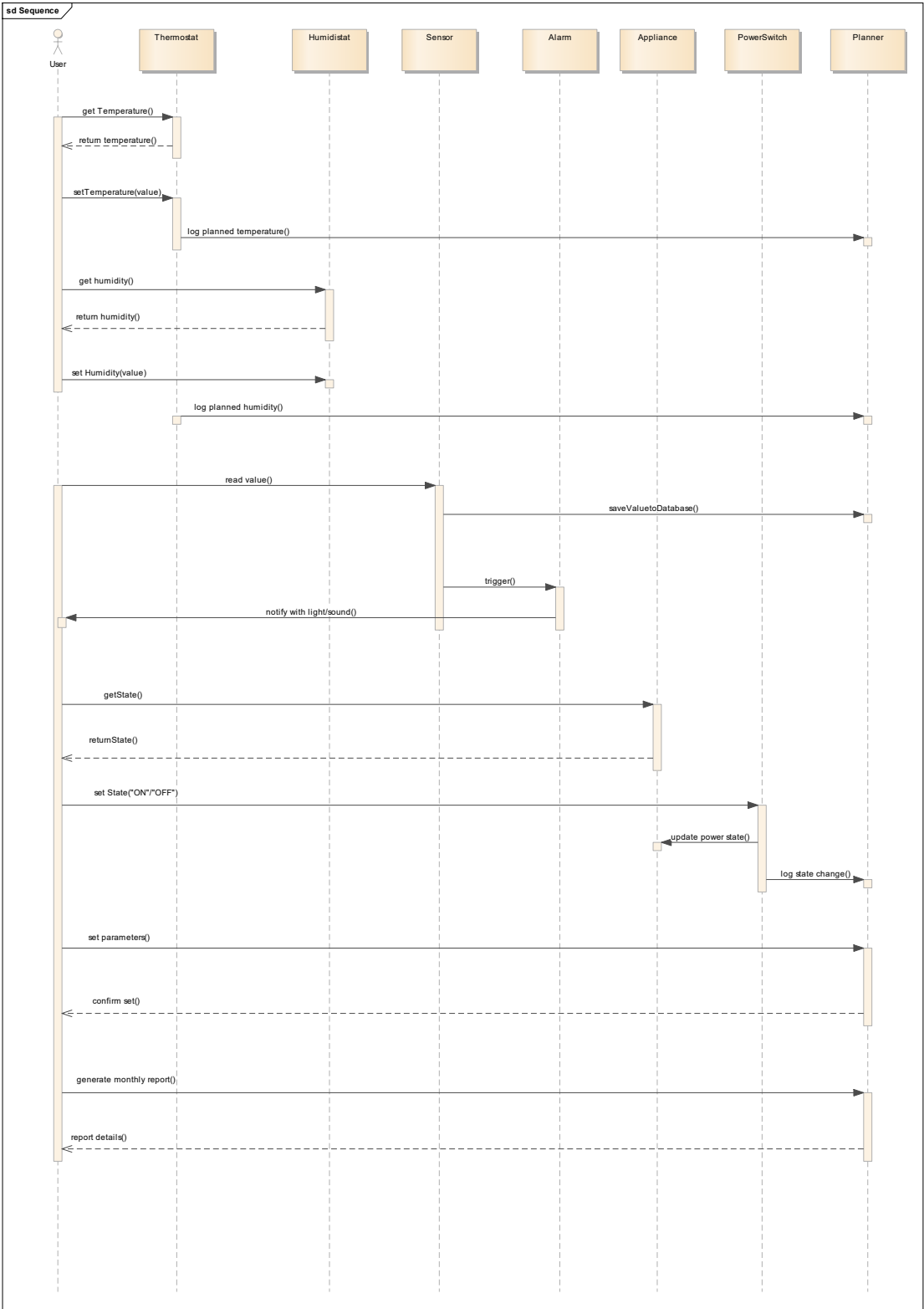
The system shall be able to allow a web-ready computer, cell phone, or PDA to control a home's temperature, humidity, lights, security, and the state of small appliances, and users shall be able to monitor and control home devices and systems. The system shall be equipped with various environmental controllers and sensors, and the sensors shall be able to read and store values in the home database. The user shall be able to read the temperature at a thermostat position and set thermostat temperatures between 60°F and 80°F in one-degree increments. If a thermostat device allows manual temperature setting, the setting shall remain in effect until the end of the planned or default time period, after which the planned or default setting shall be used for the next time period. The system shall be compatible with a centralized HVAC (Heating, Ventilation and Air Conditioning) system and shall adhere to the standards, policies, and procedures of ASHRAE 2010. The user shall be able to monitor and control home humidity from any location using a web-ready computer, cell phone, or PDA, and shall be able to read humidity at a humidistat position and set humidity levels between 30% and 60% in 1% increments. The system shall use wireless signals through a master control unit to communicate with humidistats. The system shall consist of contact sensors and security alarms and shall be able to manage up to fifty door and window contact sensors, and shall be able to activate both light and sound alarms through dedicated subsystems. The system shall be able to provide information about the state of a power

switch (OFF or ON), indicating whether an appliance connected to the switch is OFF or ON, and shall be able to change the state of a power switch (OFF to ON or ON to OFF), thereby changing the state of the connected appliance. If a user changes the power switch manually, the state shall remain in effect until the end of the planned or default time period, after which the planned or default setting shall be applied for the next time period. The system shall be able to set various preset home parameters including temperature, humidity, security contacts, and appliance/light status for defined time periods and shall provide a monthly planner. The system shall allow users to override planned parameter values through a DH website or through manual switches on household devices. The system shall be able to provide a report on the management and control of the home.

Digital Home System Class Diagram



Digital Home System Sequence Diagram



College Registration System Description

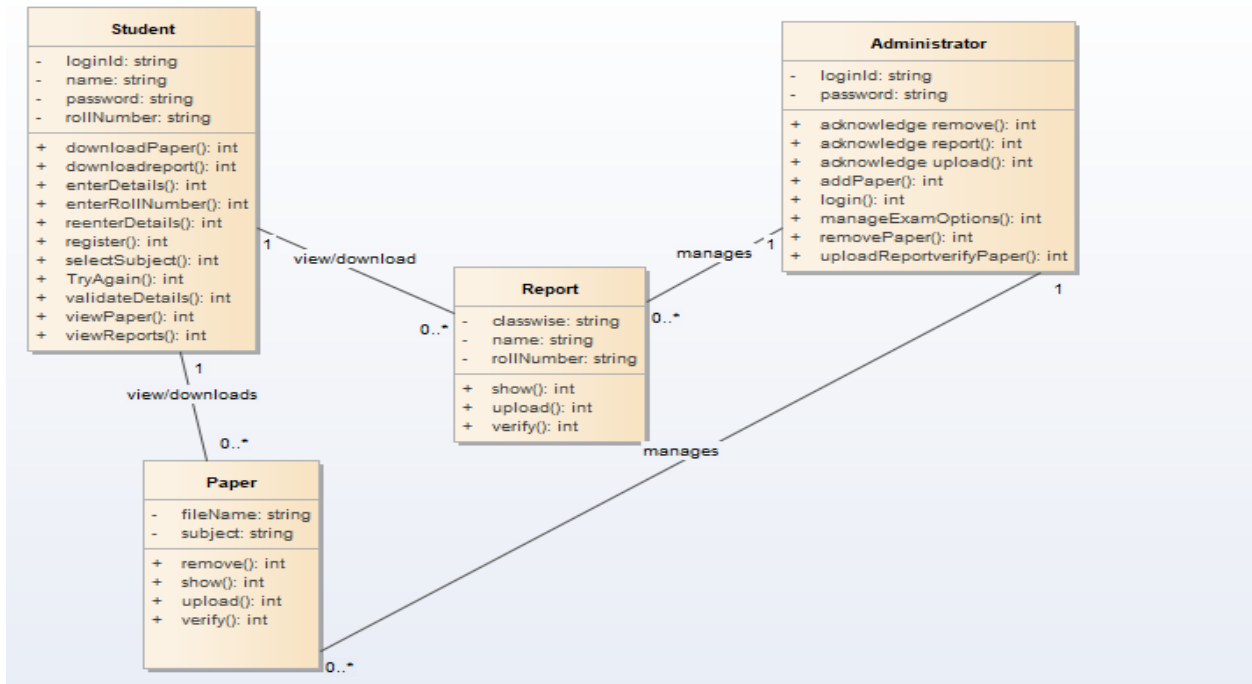
The student clicks on “Register here” button on the Home Page. The system displays the Registration page. The student enters his details. The system validates the details. If all the information is not filled by the student then the validation is not done. The system displays “Successful validation” message to the student. The user must be registered. The user enters his login Id with password. The user clicks on the confirm button. The system checks that all of the required information was entered. If the entered information is wrong then system asks user to reenter the details. The system validates the entered information against the tables stored in the database. The administrator must be logged in. The system opens a list of options. The administrator clicks on the manage exam options. From the list, He clicks on add exam papers option. The system asks him to upload the papers. The administrator uploads the paper. The system asks to verify the uploaded papers. The administrator verifies the paper, clicks on upload button. The system acknowledges by showing a message “Paper successfully uploaded”. The administrator must be logged in. The system opens a list of options. The administrator clicks on the manage exam options. From the list, He clicks on remove exam papers option. The system asks him to select the papers to be removed. The administrator selects the paper. The system asks administrator to confirm again. The administrator confirms. The system acknowledges by showing a message “Paper successfully removed”. The student must be logged in. The system opens a list of options. The student clicks on view paper option. The system asks to select the subject. If no subject is chosen then the system asks student to select a subject again. The student selects the subject of which he wants to see the paper. The system shows the paper. The student downloads the paper. The administrator must be logged in. The system opens a list of options. The administrator clicks on the Upload reports link. The system opens upload reports page. The administrator uploads the reports of the students class-wise. The system acknowledges by showing “Reports successfully uploaded” message. The student must be logged in. The system opens a list of options. The student clicks on the View reports link. The student enters his name with roll number. The system verifies it. If the information entered by student is incorrect then system asks him to try again. The system shows his report. The student downloads the report.

College Registration System (SNL)

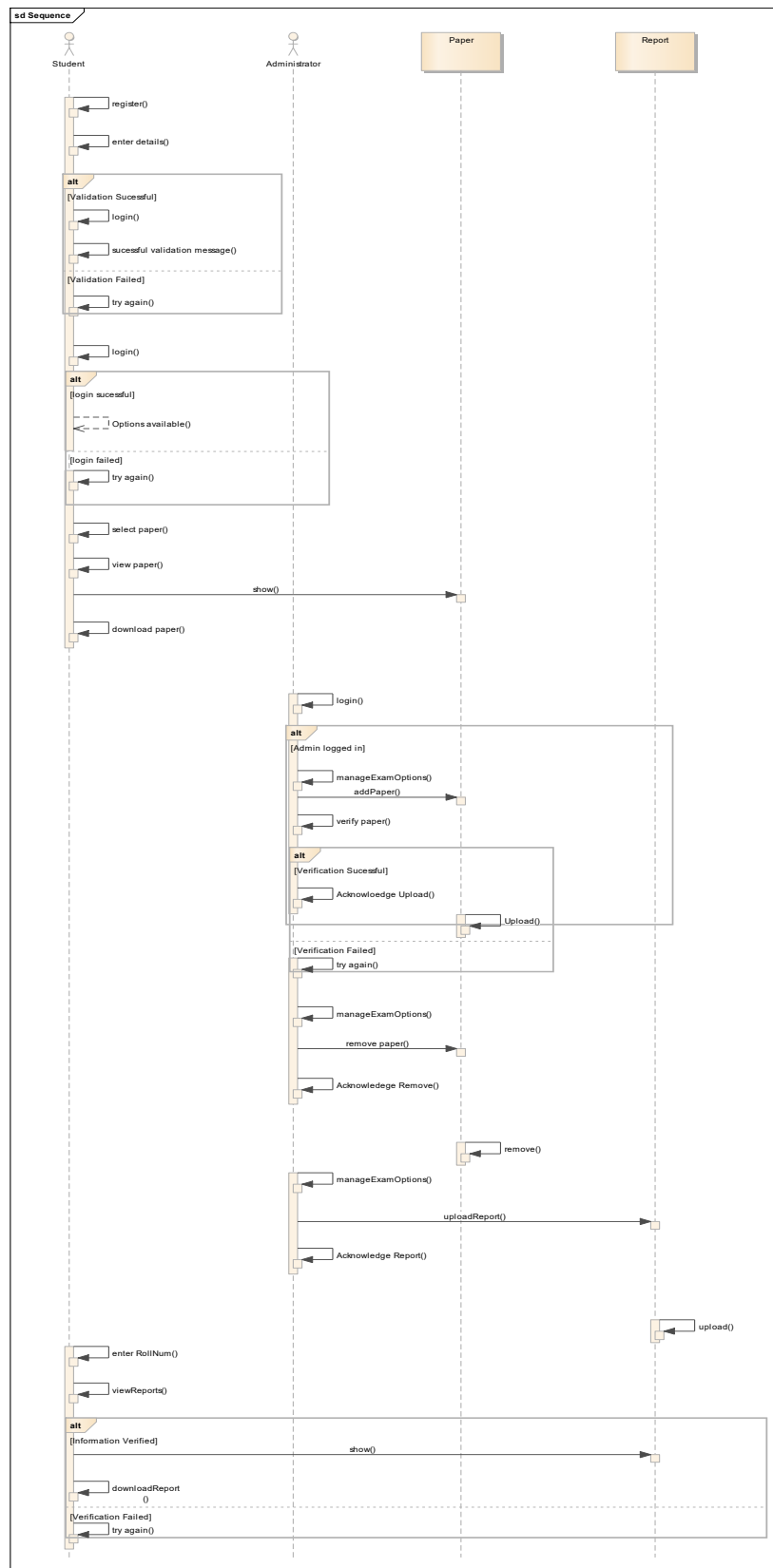
The system shall provide student with the ability to click on register here button on the home page. The system shall be able to display the registration page. The system shall provide student with the ability to enter his details. The system shall be able to validate the details. If all the information is not filled by the student then the system shall be able to show that validation is not done. The system shall be able to display successful validation message to the student. The system shall be able to ensure the user is registered. The system shall provide user with the ability to enter his login ID with password. The system shall provide user with the ability to click on the confirm button. The system shall be able to check that all of the required information entered. If the entered information is wrong then the system shall be able to ask user to reenter details. The system shall be able to validate the entered information against the tables stored in the database. The system shall be able to ensure the administrator is logged in. The system shall be able to open a list of options. The system shall provide administrator with the ability to click on the manage exam options. The system shall provide administrator with the ability to click on add exam papers option. The system shall be able to ask him to upload the papers. The system shall provide administrator with the ability to upload the paper. The system shall be able to ask to verify the uploaded papers. The system shall provide administrator with the ability to verify the paper, clicks on upload button. The system shall be able to acknowledge by showing a message

paper successfully uploaded. The system shall be able to ensure the administrator is logged in. The system shall be able to open a list of options. The system shall provide administrator with the ability to click on the manage exam options. The system shall provide administrator with the ability to click on remove exam papers option. The system shall be able to ask him to select the papers to removed. The system shall provide administrator with the ability to select the paper. The system shall be able to ask administrator to confirm again. The system shall provide administrator with the ability to confirm. The system shall be able to acknowledge by showing a message paper successfully removed. The system shall be able to ensure the student is logged in. The system shall be able to open a list of options. The system shall provide student with the ability to click on view paper option. The system shall be able to ask to select the subject. If no subject is chosen then the system shall be able to ask student to select a subject again. The system shall provide student with the ability to select the subject of which he wants to see the paper. The system shall be able to show the paper. The system shall provide student with the ability to download the paper. The system shall provide administrator with the ability to logged in. The system shall be able to open a list of options. The system shall be able to ensure the administrator is logged in. The system shall be able to open a list of options. The system shall provide administrator with the ability to click on the upload reports link. The system shall be able to open upload reports page. The system shall provide administrator with the ability to upload the reports of the students classwise. The system shall be able to acknowledge by showing reports successfully uploaded message. The system shall be able to ensure the student is logged in. The system shall be able to open a list of options. The system shall provide student with the ability to click on the view reports link. The system shall provide student with the ability to enter his name with roll number. The system shall be able to verify it. If the information entered by student is incorrect then the system shall be able to ask him to try again. The system shall be able to show his report. The system shall provide student with the ability to download the report.

College Registration System Class Diagram



College Registration System Sequence Diagram



EStore Description

The system displays all the products that can be configured. The system allows user to select the product to configure. The system displays all the available components of the product to configure. The system enables user to add one or more component to the configuration. The system notifies the user about any conflict in the current configuration. The system allows user to update the configuration to resolve conflict in the current configuration. The system allows user to confirm the completion of current configuration. The system displays detailed information of the selected products. The system provides browsing options to see product details. The system displays detailed product categorization to the user. The system enables user to enter the search text on the screen. The system enables user to select multiple options on the screen to search. The system displays all the matching products based on the search. The system displays only 10 matching results on the current screen. The system enables user to navigate between the search results. The system notifies the user when no matching product is found on the search. The system allows user to create profile and set his credential. The system authenticates user credentials to view the profile. The system allows user to update the profile information. The system displays both the active and completed order history in the customer profile. The system allows user to select the order from the order history. The system displays the detailed information about the selected order. The system displays the most frequently searched items by the user in the profile. The system allows user to register for newsletters and surveys in the profile. The system provides online help, FAQ's customer support, and sitemap options for customer support. The system allows user to select the support type he wants. The system allows user to enter the customer and product information for the support. The system displays the customer support contact numbers on the screen. The system allows user to enter the contact number for support personnel to call. The system displays the online help upon request. The system displays the FAQ's upon request. The system maintains customer email information as a required part of customer profile. The system sends an order confirmation to the user through email. The system displays detailed invoice for current order once it is confirmed. The system optionally allows user to print the invoice. The system provides shopping cart during online purchase. The system allows user to add/remove products in the shopping cart. The system displays different shipping options provided by shipping department. The system enables user to select the shipping method during payment process. The system displays the shipping charges. The system displays tentative duration for shipping. The system allows user to enter the order information for tracking. The system displays the current tracking information about the order. The system calculates tax for the order. The system displays tax information for the order. The system displays available payment methods for payment. The system allows user to select the payment method for order. The system displays the orders that are eligible to change. The system allows user to select the order to be changed. The system allows user to cancel the order. The system allows user to change shipping, payment method. The system notifies the user about any changes made to the order. The system displays the reviews and ratings of each product, when it is selected. The system enables the user to enter their reviews and ratings. The system displays all the available financing options. The system allows user to select the financing option. The system notifies the use about the financing request. The system allows user to view detailed sitemap. The system displays all the available promotions to the user. The system allows user to select available promotion. The system allows user to confirm the purchase. The system enables user to enter the payment information. The system provides a uniform look and feel between all the web pages. The system provides a digital image for each product in the product catalog. The system provides use of icons and toolbars. The system provides handicap access. The system provide multi language support. The system provides storage of all databases on redundant

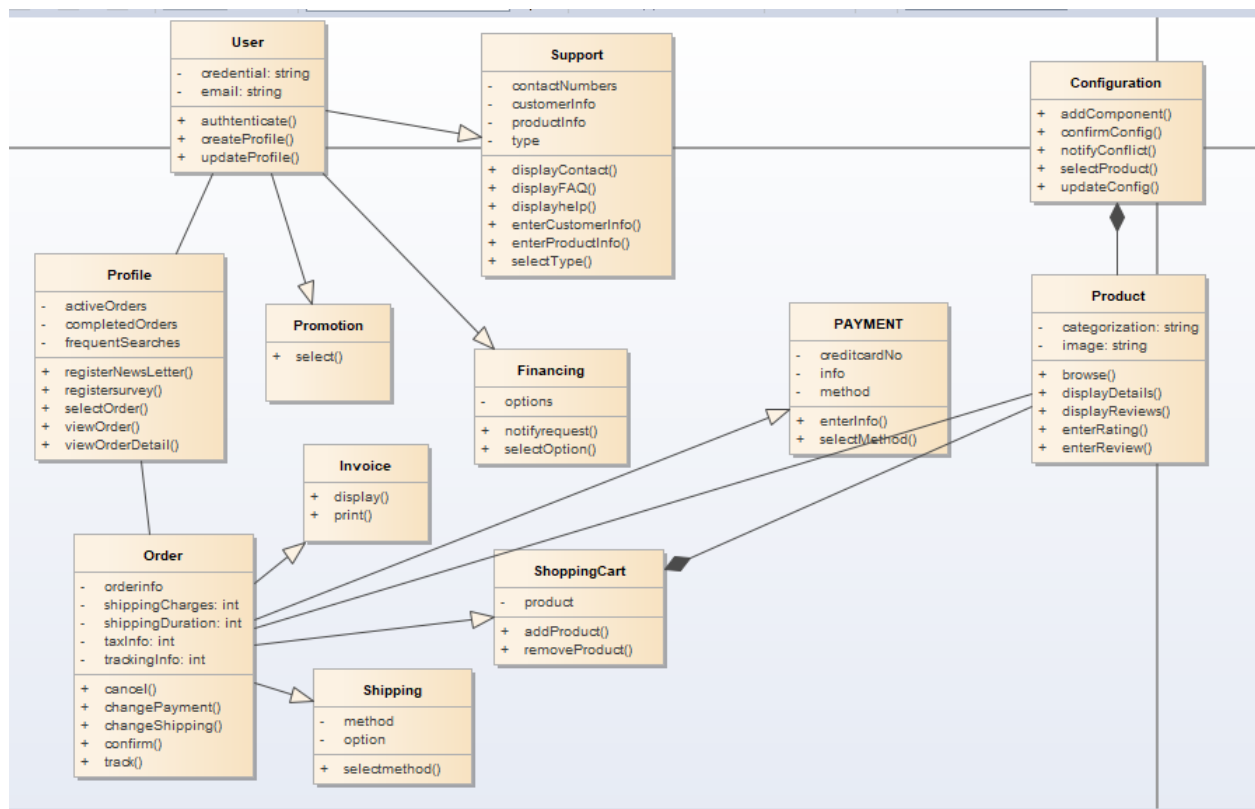
computers with automatic switchover. The system provides for replication of databases to off-site storage locations. The system provides RAID V Disk Striping on all database storage disks. The system provides a contractual agreement with an internet service provider for T3 access with 99.9999% availability. The system provides a contractual agreement with an internet service provider who can provide 99.999% availability through their network facilities onto the internet. The product is based on web and has to be run from a web server. The product takes initial load time depending on internet connection strength which also depends on the media from which the product is run. The performance depends upon hardware components of the client/customer. The system uses secure sockets in all transactions that include any confidential customer information. The system automatically logs out all customers after a period of inactivity. The system confirms all transactions with the customer's web browser. The system does not leave any cookies on the customer's computer containing the user's password. The system does not leave any cookies on the customer's computer containing any of the user's confidential information. The web browser never displays a customer's password. It always is echoed with special characters representing typed characters. The web browser never displays a customer's credit card number after retrieving from the database. It always is shown with just the last 4 digits of the credit card number. The system's back-end servers never display a customer's password. The customer password may be reset but never shown. The system backend servers are only accessible to authenticated administrators. The system backend databases are encrypted. The system is built using a standard web page development tool that conforms to either IBM's CUA standards or Microsoft's GUI standards.

EStore (SNL)

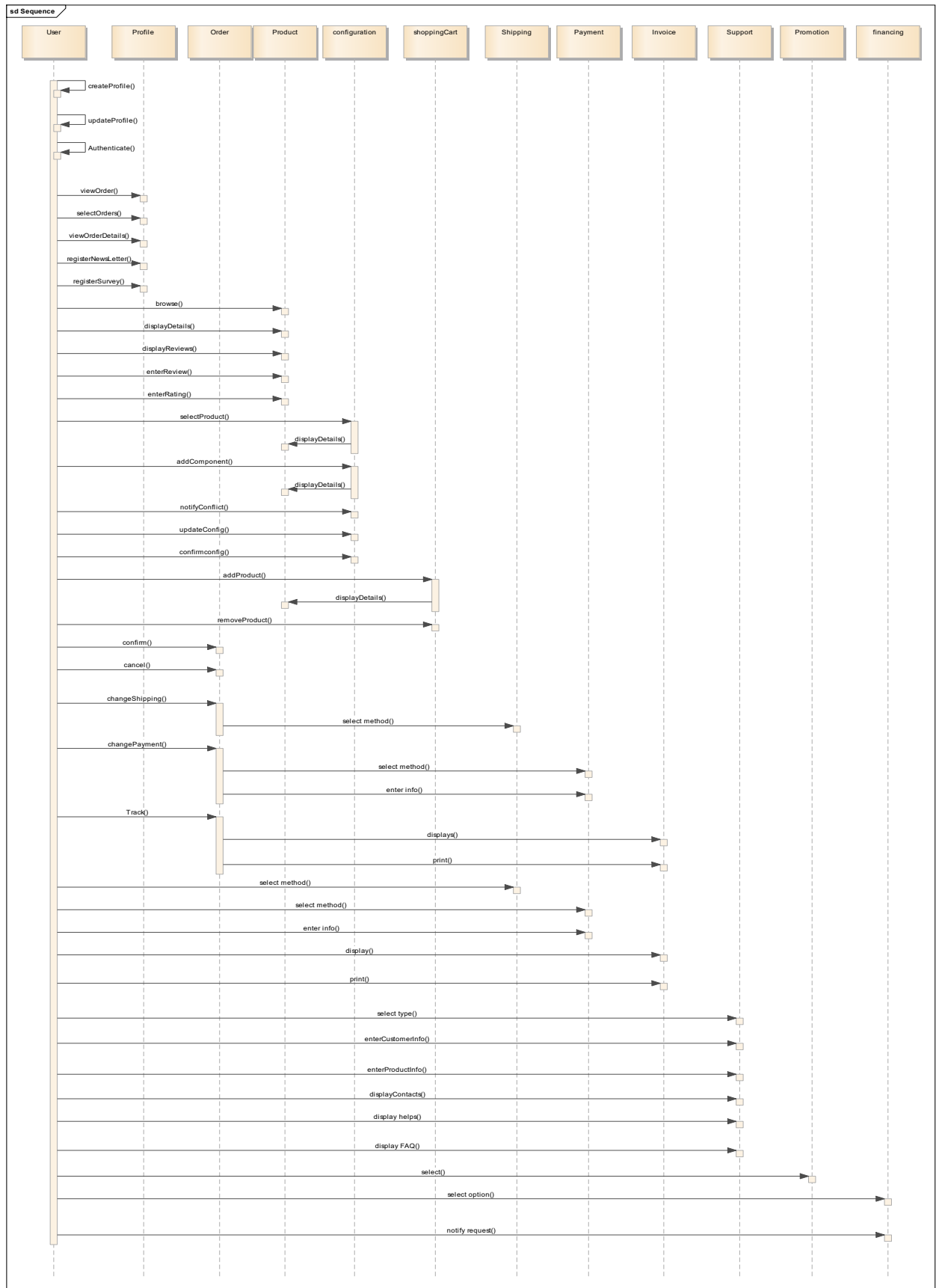
The system shall be able to display all products that can be configured and shall allow the user to select a product for configuration. The system shall display all available components of the selected product and shall enable the user to add one or more components to the configuration. The system shall notify the user about any conflict in the current configuration and shall allow the user to update the configuration to resolve conflicts, and the user shall be able to confirm completion of the configuration. The system shall display detailed information of selected products and shall provide browsing options for product details, along with detailed product categorization. The system shall enable the user to enter search text and select multiple search options, and shall display matching products with only 10 results per screen while allowing navigation between results, and shall notify the user when no matching product is found. The system shall allow the user to create a profile and set credentials, and shall authenticate user credentials to view and update the profile. The system shall display active and completed order history in the customer profile and shall allow the user to select an order to view detailed information. The system shall display frequently searched items in the profile and shall allow the user to register for newsletters and surveys. The system shall provide online help, FAQs, customer support, and sitemap options, and shall allow the user to select support type and enter customer and product information for support, while also displaying support contact numbers and allowing entry of contact numbers for support personnel. The system shall display online help and FAQs upon request. The system shall maintain customer email information as part of the profile and shall send order confirmations via email. The system shall display detailed invoices upon order confirmation and may optionally allow printing of invoices. The system shall provide a shopping cart during online purchase and shall allow the user to add or remove products. The system shall display shipping options, allow selection of shipping methods, and shall display shipping charges and estimated delivery duration. The system shall allow entry of order tracking information and shall display tracking status. The system shall calculate and display tax information for orders. The system shall display available payment methods and shall

allow selection of payment method. The system shall display orders eligible for modification and shall allow selection and cancellation of orders as well as changes to shipping and payment methods, and shall notify the user of any changes made. The system shall display product reviews and ratings and shall allow the user to enter reviews and ratings. The system shall display financing options and shall allow selection of financing options, and shall notify the user about financing requests. The system shall allow viewing of sitemap and display available promotions, and shall allow selection of promotions and confirmation of purchase. The system shall enable the user to enter payment information and shall provide a uniform look and feel across all web pages. The system shall provide digital images for products, icons, and toolbars, and shall support handicap access and multi-language support. The system shall provide redundancy in database storage with automatic switchover and replication to off-site locations, and shall use RAID storage mechanisms. The system shall provide contractual internet service with high availability (up to 99.9999%) and shall operate over a web-based server environment, with performance depending on internet connectivity and client hardware. The system shall use secure sockets for all confidential transactions and shall automatically log out inactive users. The system shall confirm all transactions through the web browser and shall ensure that no cookies store passwords or confidential information. The system shall ensure passwords are never displayed and are masked with special characters, and credit card numbers are shown only with the last four digits visible, while backend servers shall never display passwords.

EStore Class Diagram



EStore Sequence Diagram



iCoot System Description

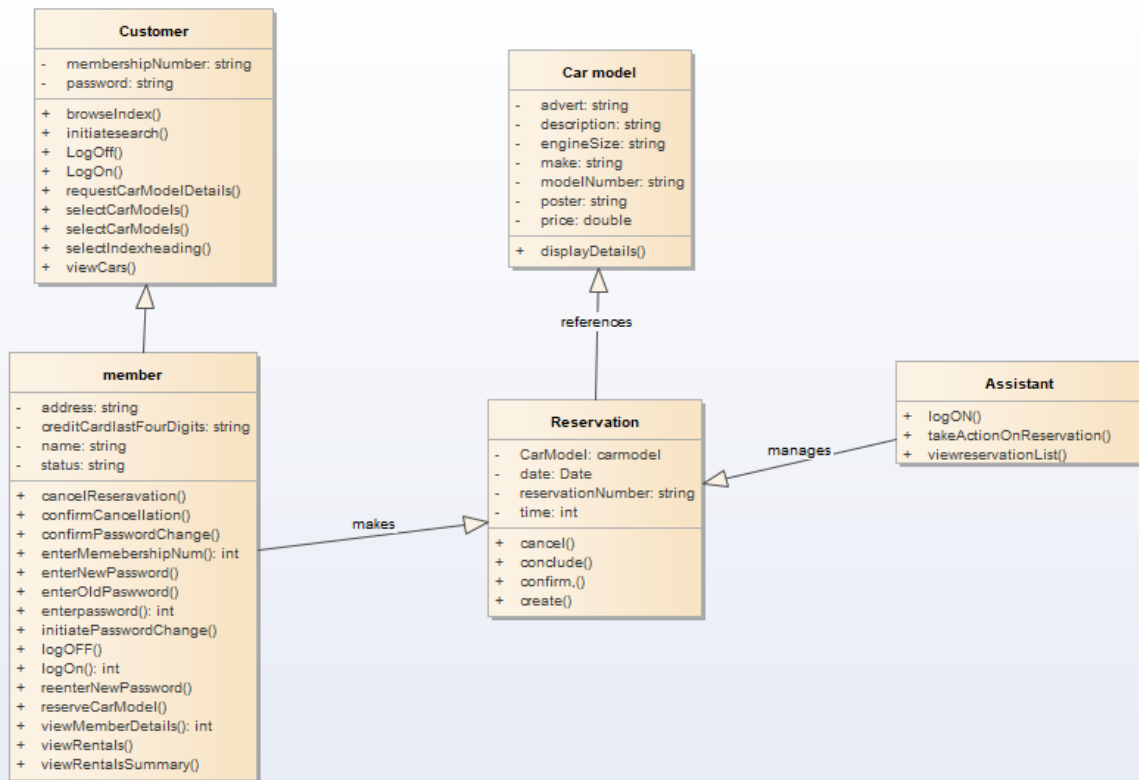
The Customer browses the index. He selects an index heading. The Customer selects to view Car-Models for the selected index heading. To view the results ViewResults usecase is carried out. The system presents the Customer with a summary of each retrieved CarModel, including model number with price. The customer views the cars. To view the car model details, ViewCarModelDetails use case is carried out. The customer should be logged into iCoot system. The Customer selects one of the matching CarModels. The Customer requests details of the selected CarModel. The system displays details for the selected car model including makes, engine size, price, description, advert, poster. If Customer wants to make a reservation then MakeReservation use case is carried out. The customer selects the required categories and he selects the required Makes. The Customer also selects the required engine sizes and he initiates the search. If Customer specifies no categories, makes or engine sizes then the system ask the Customer to enter either required categories. Member has already obtained a password from their local Store. The Member enters the membership number. The Member enters the password. The Member elects to log on. The system displays the Member's Window. If the Member elects View Member Details then View_MemberDetails usecase is carried out. The system again displays the Member's Window. If the Member elects View Rentals then ViewRentals usecase is visited. The system displays a message on Member's Window. If the member elects Log Off then LogOff use case is used. The customer should be logged into the system. If the customer is not logged in then the system asks him to log in, LogOn usecase is carried out. For security reasons, the system must display only the last four digits of the Member's CreditCard number. The system presents the Member with membership details including name, address, status, amount owing, CreditCard details. The system informs the Member to verify the details. If there are corrections in the details then system asks member to contact local Store. The system verifies the details entered by the member. The system shows the details. The member is logged on in advance. The Member elects to reserve the CarModel for the details on display. The Assistant asks the Member for confirmation. The system issues a warning that failure to collect a reserved CarModel will result in a fine. If Member declines Reservation conditions then the system again asks the Member for confirmation. The member confirms the Reservation. The system shows the member the Reservation number. The system indicates the Member that an Assistant will be in touch when the requested Car is available. When an Assistant logs on to iCoot then the system gives the Assistant a list of Reservations that require action. The Assistant takes necessary action to progress Reservations. The Member elects to view the Rentals. The system presents the Member with summary of each Car the Member currently have out for rent including number plate ,due date. The system presents Member with summaries of Cars currently rented. If no information is shown then the system asks member to select a valid options. The member should be logged into iCoot system. The system asks him to enter the old password. Member enters the old password. The system asks member to enter a new password. The Member enters a new password. The Member enters the new password again. The Member initiates the change. The system asks the Member for confirmation. The member confirms it. The system validates the password. The system displays a warning to the Member "Please memorize your password". The member should be logged into iCoot system. If the customer is not logged in then the system asks him to log in, LogOn usecase is carried out. He should have a reservation to view the reservations. The system shows a list of reservation made by the member. The system displays the validated summaries of member's reservations such as date, time, car models etc. The Member should be logged into iCoot system and should already have a Reservation. The Member selects to cancel the Reservation. The Assistant asks the Member for confirmation. Member confirms to cancel the Reservation. If Member doesn't confirm a cancellation then that reservation is not cancelled by the system. The system marks the Reservation as Concluded. The system updates the Assistant's terminals accordingly. The member should be logged into iCoot system. If the customer is not logged in then the system asks him to log in, LogOn usecase is carried out. The member selects to log off the iCoot system. The system ends the current session. The system makes the Member functions unavailable to Member. If the member does not interact with the system for ten minutes then the member is logged off automatically by the system. The system validates the member's log off. The system asks customer to look for the CarModels. The customer retrieves a subset of CarModels from the catalog. The system displays all the possible models on the options selected by the customer.

iCoot (SNL)

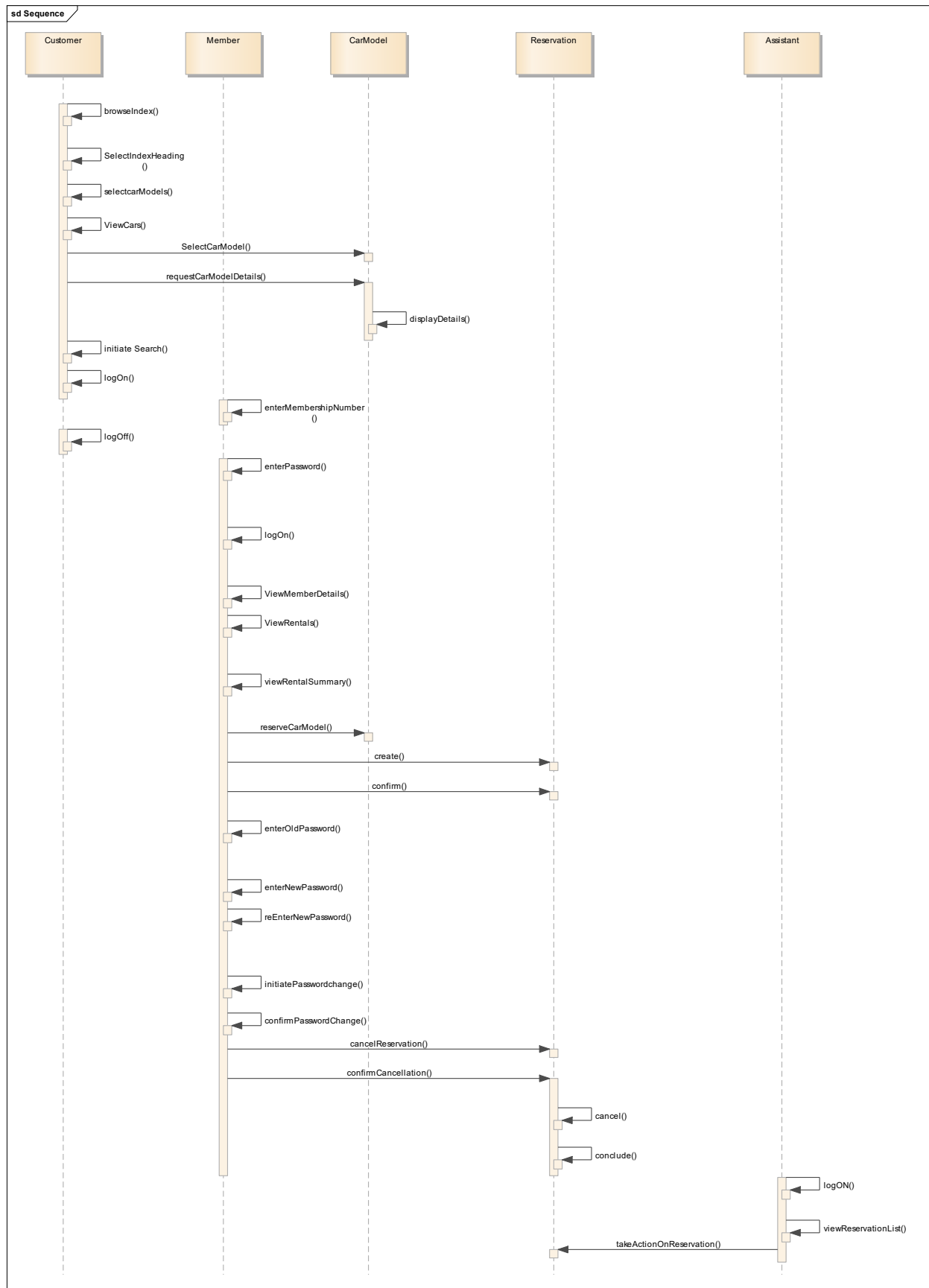
The system shall provide customer with the ability to browse the index. The system shall provide customer with the ability to select index heading. The system shall provide customer with the ability to select to view Car-Models for the selected index heading. The system shall be able to carry out the ViewResults use case when viewing results. The system shall be able to present the Customer with a summary of each retrieved CarModel, including model number with price. The system shall provide customer with the ability to view the cars. The system shall provide customer with the ability to logged in. The system shall provide customer with the ability to

select one of the matching CarModels. The system shall provide customer with the ability to request details of the selected CarModel. The system shall be able to display details for the selected car model including makes, engine size, price, description, advert, poster. If Customer wants to make a reservation then the system shall be able to carry MakeReservation use case out. The system shall provide customer with the ability to select the required categories. The system shall provide customer with the ability to select makes. The system shall provide customer with the ability to also select the required engine sizes and he initiates the search. If Customer specifies no categories, makes or engine sizes then the system ask the Customer to enter either required categories. The system shall provide member with the ability to already obtain a password from their local Store. The system shall provide member with the ability to enter the membership number. The system shall provide member with the ability to enter the password. The system shall provide member with the ability to elect to log on. The system shall be able to display the Members Window. If the Member elects View Member Details then View_MemberDetails use case is carried out. The system shall be able to again display the Members Window. If the Member elects View Rentals then ViewRentals use case is visited. The system shall be able to display a message on Members Window. If the member elects Log Off then LogOff use case is used. The system shall be able to ensure the customer is logged into the system. The system shall provide customer with the ability to enter shipping address. The system shall be able to present shipping options. The system shall provide customer with the ability to select shipping option. The system shall be able to ask which credit/debit card used. The system shall provide customer with the ability to select to use the credit card which is stored in the system. If customer wants to use another card for payment then the system shall be able to enter customer details of that card. The system shall be able to ask for final confirmation to place an order. The system shall provide customer with the ability to place an order. The system shall be able to return confirmation number. The system shall be able to exist delivery order. The system shall provide supplier with the ability to need to log in, login usecase carried out. The system shall be able to display the delivery order made by customer to supplier. The system shall provide supplier with the ability to check the order. The system shall provide items with the ability to determine that available in inventory to satisfy delivery order. If item is out of stock then the system shall be able to reject supplier delivery of that order by asking to select another supplier. The system shall be able to reserve the selected items for a customer. The system shall be able to display the inventory information to supplier. The system shall be able to confirm that the items reserved to delivered. The system shall be able to ask customer to look for the CarModels. The system shall provide customer with the ability to retrieve a subset of CarModels from the catalog. The system shall be able to display all the possible models on the options selected by the customer. The system shall be able to display only the last four digits of the Member's CreditCard number. The system shall be able to present the member with membership details including name, address, status, amount owing, credit card details. The system shall be able to inform the member to verify the details. If there are corrections in the details then the system shall be able to ask member to contact local Store. The system shall be able to verify the details entered by the member. The system shall be able to show the details. The system shall be able to ensure the member is logged into the system. The system shall be able to ensure the member is logged into the system. If the customer is not logged in then the system shall be able to ask him to log in, LogOn usecase is carried out. The system shall be able to ensure the customer is logged into the system. he system shall be able to ask customer to look for the CarModels. The system shall provide customer with the ability to retrieve a subset of CarModels from the catalog.

iCoot System Class Diagram



iCoot System Sequence Diagram



Online Bus Reservation Description

The Customer clicks the log-in button on the Home Page. The system displays the log-in page. The customer enters his login Id with password. The customer clicks on the confirm button. The system checks that all of the required information was entered. If the entered information is wrong then system asks customer to reenter the details. The system validates the entered information against the tables stored in the database. The admin clicks the log-in button on the Home Page. The system displays the log-in page. The admin enters his login Id with password. The admin clicks on the confirm button. The system checks that all of the required information was entered. If the entered information is wrong then system asks admin to reenter the details. The system validates the entered information against the tables stored in the database. The customer already has navigated to the main options screen. The customer enters the details of source station, destination station, date to get the best possible buses. The customer also enters the type of bus he wants to book. Customer enters all the information- name, age, number of passengers, source station, destination station. The customer can use help facility to know how to fill the information, HelpFacility use case is carried out. He clicks the "Search" button. If all the required information is not entered by the customer then the system highlights the blank fields to be entered. After searching, the System displays a list of all the buses that matches the customer's input, each bus in the list has an associated price. Customer selects the "Purchase Ticket" link from the list of options. The system displays a screen with an input field for a reservation number. The customer enters a reservation number. He clicks the "Submit" button. System displays the details of the reservation and the customer selects the "Purchase Ticket" option. System displays a screen with input fields for: a credit card number, cardholder name, credit card expiration date. Customer enters the details for the credit card number, the cardholder name, the credit card expiration date and he clicks the "Submit" button. If customer is stuck in the process then he can use help facility to know how to fill the information, HelpFacility use case is carried out. System displays the success message. A reservation has already been made and customer must have successfully navigated to the main options screen. Customer Selects the "Cancel Reservation" option. System displays a screen with an input field for a reservation number and Customer enters a reservation number and he clicks the "Submit" button. The customer can use help facility, HelpFacility use case is carried out. System displays the details of the reservation. Customer Selects the "Process Cancellation" option and then system displays the main options screen again. The website provides instructions to the customers on how to book tickets, search buses, cancel reservation. The customer clicks on help facility. The system displays a list of facilities. The customer clicks on the type of help required. The system shows the instructions. The admin manages the bus. The system asks Admin whether he wants to Add Bus or Remove Bus. The Admin selects to add bus, AddBus use case is carried out. If Admin selects to remove bus, then RemoveBus use case is carried out. The Admin already has navigated to the main options screen. The system validates the log in information. The admin selects on adding the bus. The system asks admin to fill all the required information for adding a bus in the database. The admin fills the information. The system displays a success message on the screen. If all the required information is not filled by Admin then the system shows an error message. The Admin already has navigated to the main options screen. The system validates the log in information. The admin selects on remove bus. The system asks admin to fill all the required information for removing a bus in the database. The admin fills all the required information. The system displays a success message on the screen. If all the required information is not filled by Admin then the system shows an error message.

Online Bus Reservation System (SNL)

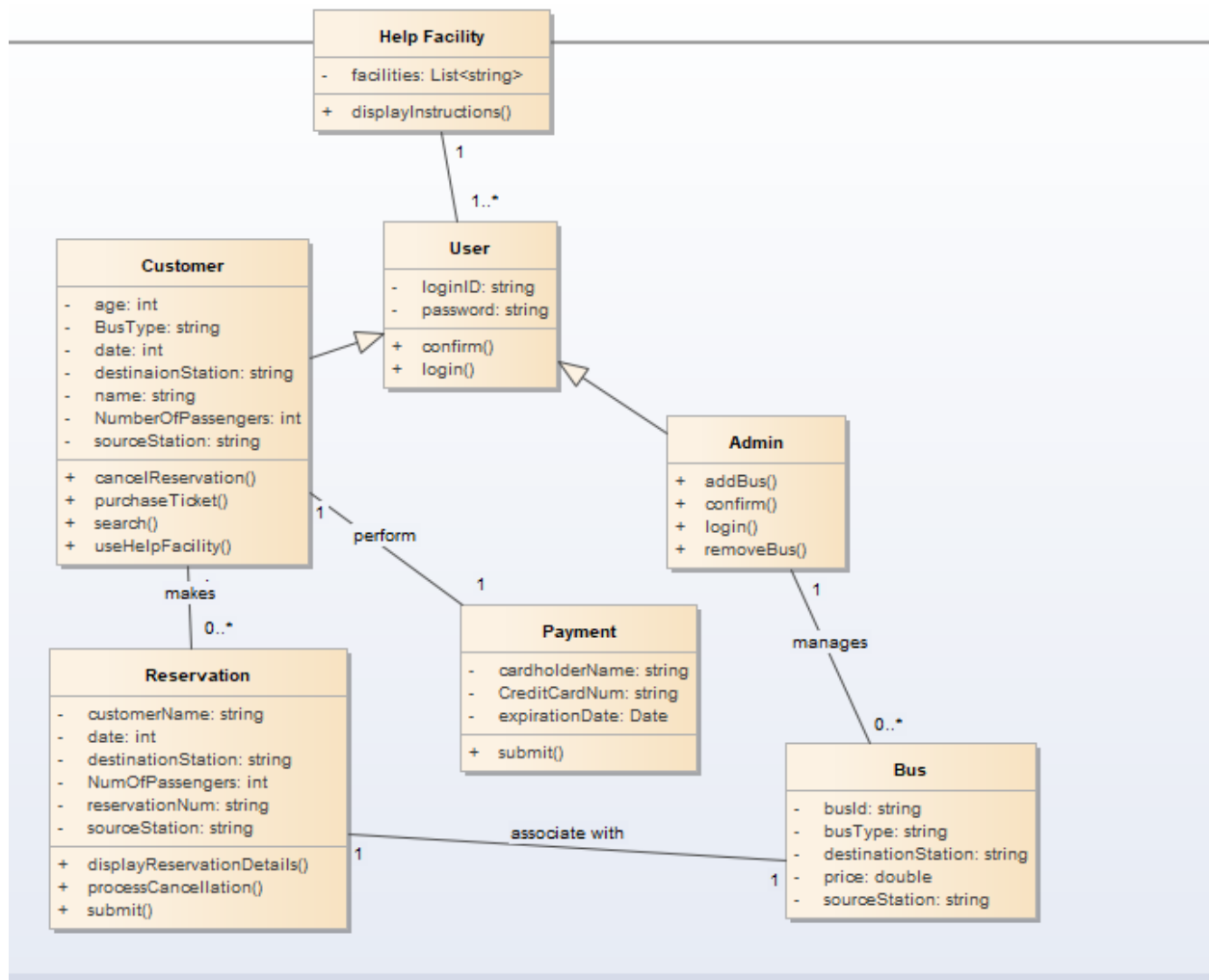
The system shall provide customer with the ability to click the login button on the home page. The system shall be able to display the login page. The system shall provide customer with the ability to enter his login ID with password. The system shall provide customer with the ability to click on the confirm button. The system shall be able to check that all of the required information entered. If the entered information is wrong then the system shall be able to ask customer to reenter details. The system shall be able to validate the entered information against the tables stored in the database. The system shall provide admin with the ability to click the login button on the home page. The system shall be able to display the login page. The system shall provide admin with the ability to enter his login ID with password. The system shall provide admin with the ability to click on the confirm button. The system shall be able to check that all of the required information entered. If the entered information is wrong then the system shall be able to ask admin to reenter details. The system shall be able to validate the entered information against the tables stored in the database. The system shall provide customer with the ability to already

navigate to the main options screen. The system shall provide customer with the ability to enter the details of source station, destination station, date to get the best possible buses. The system shall provide customer with the ability to also enter the type of bus he wants to book. The system shall provide customer with the ability to enter all the information name, age, number of passengers, source station, destination station. The system shall provide customer with the ability to use help facility to know how to fill the information, HelpFacility use case carried out. The system shall provide customer with the ability to click the search button.

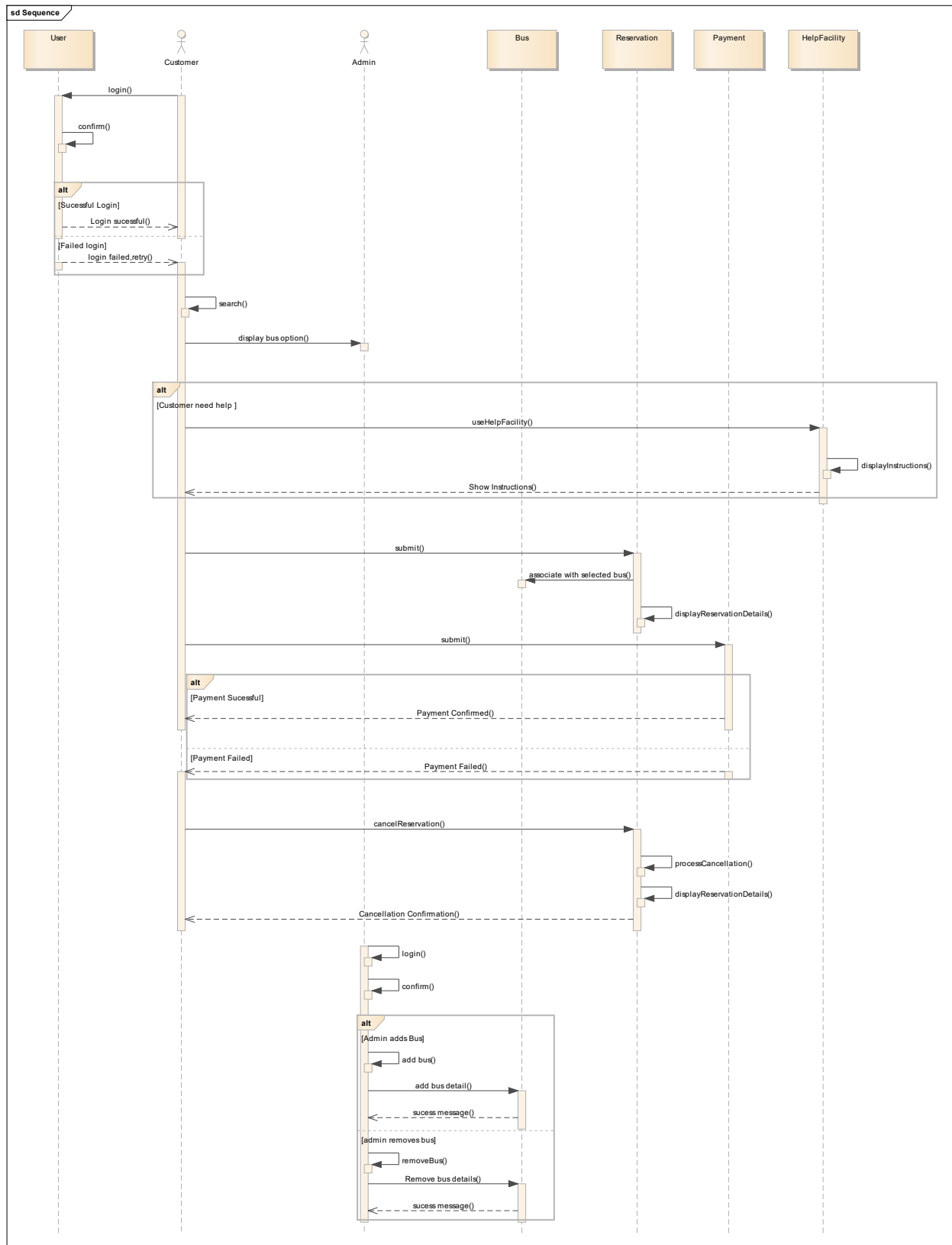
The system shall be able to highlight the blank fields to be entered if all the required information is not entered by the customer. The system shall be able to display a list of all the buses that matches the customers input, each in the list has an associated price. The system shall provide customer with the ability to select the purchase ticket link from the list of options. The system shall be able to display a screen with an input field for a reservation number. The system shall provide customer with the ability to enter a reservation number. The system shall provide customer with the ability to click the submit button. The system shall be able to display the details of the reservation. The system shall provide customer with the ability to select the purchase ticket option. The system shall be able to display a screen with input fields for a credit card number, cardholder name, credit card expiration date. The system shall provide customer with the ability to enter the details for the credit card number, the cardholder name, the credit card expiration date and click the submit button. If customer is stuck in the process then the system shall provide customer with the ability to use help facility to know how to fill the information, HelpFacility use case is carried out. The system shall be able to display the success message. The system shall be able to acknowledge that a reservation has already been made. The system shall provide customer with the ability to successfully navigate to the main options screen. The system shall provide customer with the ability to select the cancel reservation option. The system shall be able to display a screen with an input field for a reservation number. The system shall provide customer with the ability to use help facility, HelpFacility use case carried out. The system shall be able to display the details of the reservation. The system shall provide website with the ability to provide instructions to the customers on how to book tickets, search buses, cancel reservation. The system shall provide customer with the ability to click on help facility.

The system shall be able to display a list of facilities. The system shall provide customer with the ability to click on the type of heal required. The system shall be able to show the instructions. The system shall provide admin with the ability to manage the bus. The system shall be able to ask admin whether he wants to add bus or remove bus. The system shall provide admin with the ability to select to add bus, AddBus use case carried out. If admin selects to remove bus, then the system shall be able to carry out the RemoveBus use case. The system shall provide admin with the ability to already navigate to the main options screen. The system shall be able to validate the log in information. The system shall provide admin with the ability to select on adding the bus. The system shall be able to ask admin to fill all the required information for adding a bus in the database. The system shall provide admin with the ability to fill the information. The system shall be able to display a success message on the screen. If all the required information is not filled by admin then the system shall be able to show an error message. The system shall provide admin with the ability to fill all the required information. The system shall be able to display a success message on the screen. If all the required information is not filled by admin then the system shall be able to show an error message.

Online Bus Reservation Class Diagram



Online Bus Reservation Sequence Diagram



Online Discussion Group Description

The leader logs into the system. The system authenticates the leader to start a session. If the system fails to validate the leader then the system shows an error message. The leader chooses to create a new group. The system guides the leader to fill in required information to create the discussion group. If the system fails to create a new group then the system asks leader to check the information. The leader invites other students to join the discussion group by giving invitation message. If system fails to deliver invitation message then the leader is asked to retry the process. The system acknowledges that a new discussion group is created. There should be at least one existing discussion group. The student logs in. The system authenticates the student. The system starts the session. If the system fails to validate the student then the system shows an error message. The student chooses to join the existing discussion group. The system guides the student to fill in required information to join the discussion group. If the information filled by the student is incomplete then the system shows an error message. The system acknowledges that the student has successfully joined the discussion group. The student should have joined at least one discussion group. The student logs into the system. The system authenticates the student to start a session. If the system fails to validate the student then the system shows an error message. The student chooses to leave the discussion group. The student is guided by the system to specify the discussion group he wants to leave. The system acknowledges that the student has successfully left that group. If the student is unable to leave the selected discussion group then the system asks the student to retry. The student logs into the system. The system validate the student to start a session. If the system fails to validate the student then the system asks student to retry. The student chooses to add a presentation. The system guides the student to fill in required information to add a new presentation. The system acknowledges that the presentation is added successfully. If the presentation is not added successfully then the system shows an error message asking to retry. The student has to be a member of at least one discussion group. The student logs into the system. The system authenticates the student to start a session. If the system fails to validate the student then the system asks student to enter login id with password again. The student chooses to edit a presentation. The student edits the presentation. The system asks student to save the changes made by him. If the student selects not to save the changes then no changes in the presentation is saved. The system acknowledges that the presentation has been saved. The leader must be logged in. The leader logs into the system. The system authenticates the leader to start a session. If the user is not registered as a leader then an error message is shown by the system. Only leader is allowed to delete a presentation. The system guides the leader to delete a presentation. The system acknowledges that the presentation has been deleted. The student has to be a registered student. The student logs into the system. The system authenticates the student to start a session. If the system fails to validate the student then the system asks student to retry. The student chooses to add a comment. The system guides the student to fill in required information to add a new comment. The system acknowledges that the comment is added successfully. If the comment is not added successfully then the system shows an error message asking to retry. The student has to be a member of at least one discussion group. The student logs into the system. The system authenticates the student to start a session. If the system fails to validate the student then the system asks student to enter login id with password again. The student chooses to edit a comment made by him. If the student tries to edit a comment made by someone else then the system shows an error message. The student edits the comment. The system asks student to save the changes made by him. If the student selects not to save the changes then no changes in the comment is saved. The system acknowledges that the new comment has been saved. The leader must be logged in. The leader logs into the system. The system authenticates the leader to start a session. If the user is not registered as a leader then an error message is shown by the system. Only leader is allowed to delete a comment. The system guides the leader to delete a comment. The system acknowledges that the comment has been deleted.

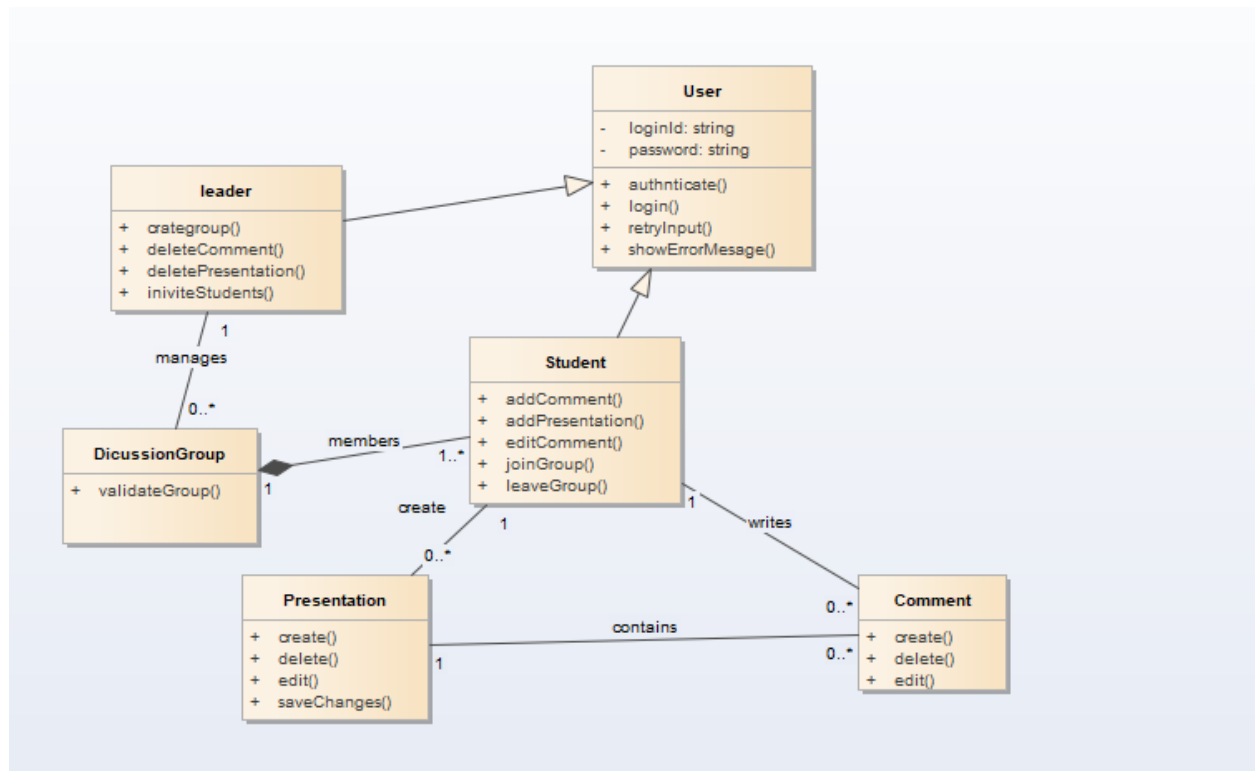
Online Discussion Group (SNL)

The system shall provide leader with the ability to log in to the system. The system shall be able to authenticate the leader to start a session. If the system fails to validate the leader then the system shall be able to show an error message. The system shall provide leader with the ability to choose to create a new group. The system shall be able to guide the leader to fill in required information to create the discussion group. If the system fails to create a new group then the system shall be able to ask leader to check information. The system shall provide leader with the ability to invite other students to join the discussion group by giving invitation message. If system fails to deliver invitation message then the system shall be able to ask leader to retry process. The system shall be able to acknowledge that a new discussion group is created. The system shall be able to have at least one existing discussion group. The system shall provide student with the ability to log in. The system shall be able to authenticate the student. The system

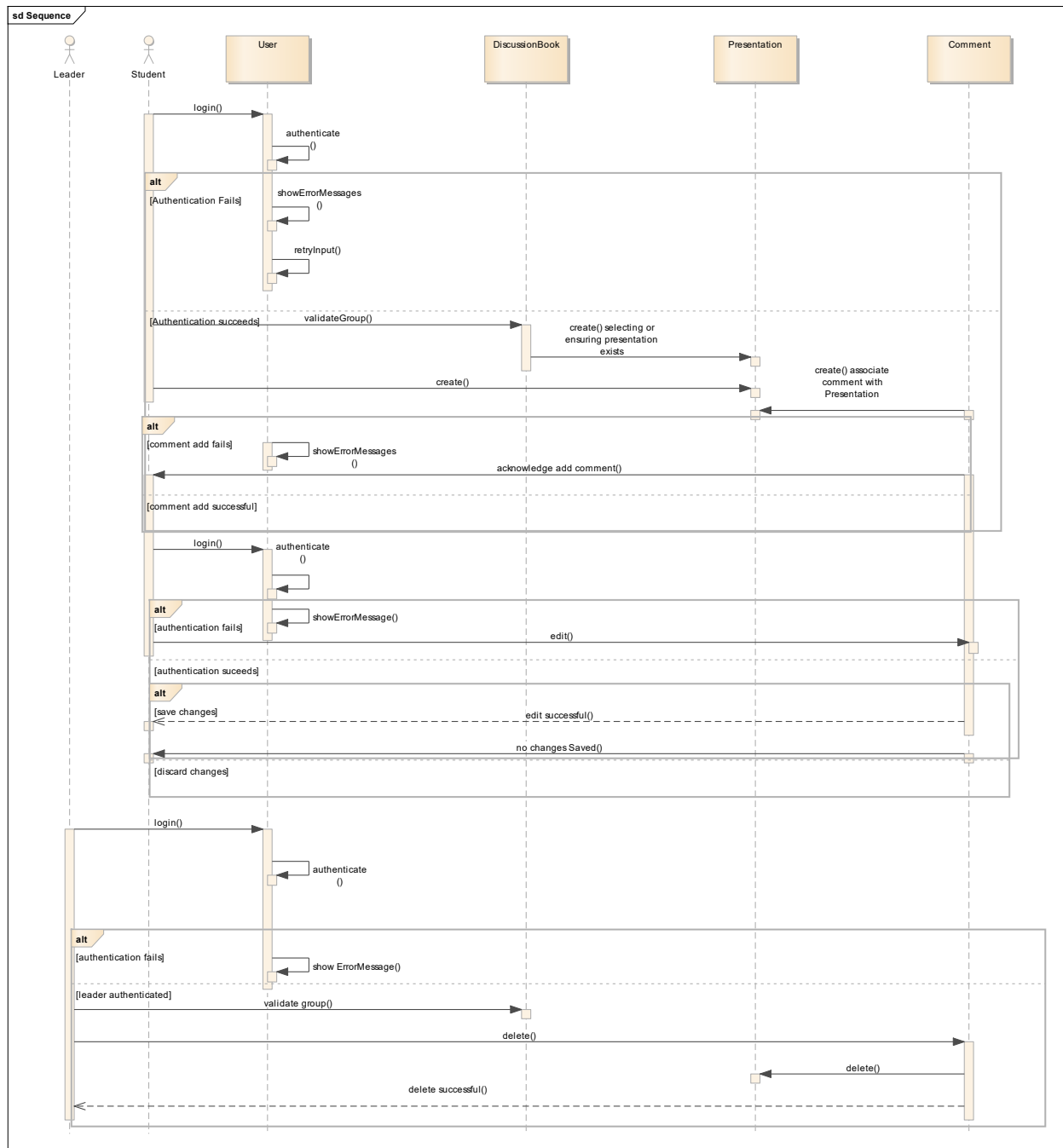
shall be able to start the session. If the system fails to validate the student then the system shall be able to show an error message. The system shall provide student with the ability to choose to join the existing discussion group. The system shall be able to guide the student to fill in required information to join the discussion group. If the information filled by the student is incomplete then the system shall be able to show an error message. The system shall be able to acknowledge that the student has successfully joined the discussion group. The system shall be able to verify that the student has joined at least one discussion group. The system shall provide student with the ability to log in to the system. The system shall be able to authenticate the student to start a session. If the system fails to validate the student then the system shall be able to show an error message. The system shall provide student with the ability to choose to leave the discussion group. The system shall provide student with the ability to specify the discussion group to leave. The system shall be able to acknowledge that the student has successfully left that group. If the student is unable to leave the selected discussion group then the system shall be able to ask student to retry. The system shall provide student with the ability to log in to the system. The system shall be able to validate the student to start a session. If the system fails to validate the student then the system shall be able to ask student to retry. The system shall provide student with the ability to choose to add a presentation. The system shall be able to guide the student to fill in required information to add a new presentation. The system shall be able to acknowledge that the presentation added successfully. If the presentation is not added successfully then the system shall be able to show an error message asking to retry. The system shall be able to verify that the student is a member of at least one discussion group. The system shall provide student with the ability to log in to the system. The system shall be able to authenticate the student to start a session. If the system fails to validate the student then the system shall be able to ask student to enter login ID with password again. The system shall provide student with the ability to choose to edit a presentation. The system shall provide student with the ability to edit the presentation. The system shall be able to ask student to save the changes made by him. If the student selects not to save the changes then the system shall be able to save no changes in presentation saved. The system shall be able to acknowledge that the presentation has been saved. The system shall be able to verify that the leader is logged in. The system shall provide leader with the ability to log in to the system. The system shall be able to authenticate the leader to start a session. If the user is not registered as a leader then the system shall be able to show an error message by. The system shall be able to allow only the leader to delete a presentation. The system shall be able to guide the leader to delete a presentation. The system shall be able to acknowledge that the presentation deleted. The system shall be able to verify that the student is a registered student. The system shall provide student with the ability to log in to the system. The system shall be able to authenticate the student to start a session. If the system fails to validate the student then the system shall be able to ask student to retry. The system shall provide student with the ability to choose to add a comment. The system shall be able to guide the student to fill in required information to add a new comment. The system shall be able to acknowledge that the comment added successfully. If the comment is not added successfully then the system shall be able to show an error message asking to retry. The system shall provide student with the ability to have to a member of at least one discussion group. The system shall provide student with the ability to log in to the system. The system shall be able to authenticate the student to start a session. The system shall be able to verify that the student is a member of at least one discussion group. The system shall provide student with the ability to log in to the system. The system shall be able to authenticate the student to start a session. If the system fails to validate the student then the system shall be able to ask student to enter login ID with password again. The system shall provide student with the ability to choose to edit a comment made by him. If the student tries to edit a comment made by someone else then the system shall be able to show an error message. The system shall provide student with the ability to edit the comment. The system shall

be able to ask student to save the changes made by him. If the student selects not to save the changes then the system shall be able to save no changes in the comment. The system shall be able to acknowledge that the new comment has been saved. The system shall be able to verify that the leader is logged in. The system shall provide leader with the ability to log in to the system. The system shall be able to authenticate the leader to start a session. If the user is not registered as a leader then the system shall be able to show an error message. The system shall be able to allow only leader to delete a comment. The system shall be able to guide the leader to delete a comment. The system shall be able to acknowledge that the comment has been deleted.

Online Discussion Group Class Diagram



Online Discussion Group Sequence Diagram



Online Pet Store Description

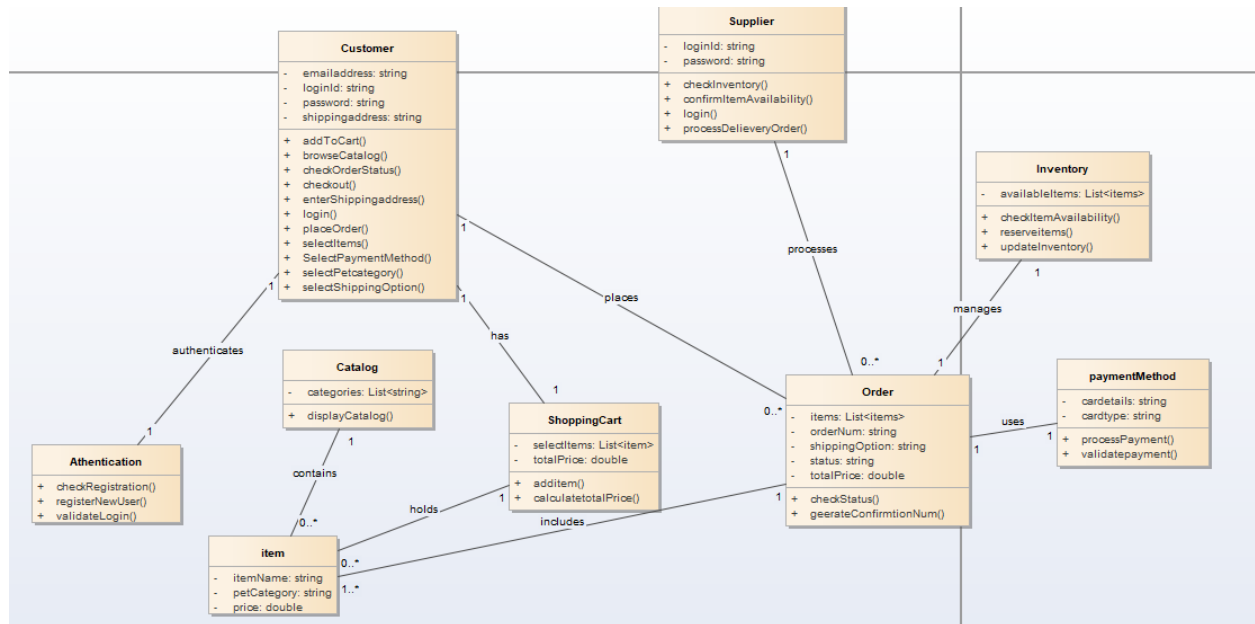
The Customer clicks the log-in button on the Home Page. The system displays the log-in page. The customer enters his login Id with password. The customer clicks on the confirm button. The system checks that all of the required information was entered. If the entered information is wrong then system asks customer to reenter the details. The system validates the entered information against the tables stored in the database. Customer browser has to be linked to supplier catalog Web site before the starting of the process. The Customer requests to browse catalog. System displays login page, Log-in usecase is carried out. If the customer is not registered then the system asks customer to get registered. Customer enters email address with a password. System confirms correct login. If the entered email address with password is incorrect then system shows login error message. The system displays catalog information to customer. The catalog consists of the categories of pets- dogs, cats , birds. The system asks user to select the pet category. The customer selects the pet category, browses items. The customer selects the items for Pet category from the catalog. If Customer does not select any Pet category from the displayed catalog then the system asks customer to browse related items for pets from the catalog. The system displays a list of items with price. Customer has to be logged in before placing an order, Log-in usecase is carried out. From the list of items for Pets from catalog, customer selects the items for his Pet. System presents detailed information about the items. Customer adds the selected item to a shopping cart. Shopping cart contents is presented to the customer. Customer selects proceed to checkout option. Customer enters shipping address. System presents shipping options. The customer selects shipping option. System asks which credit/debit card will be used. Customer selects to use the credit card which is stored in the system. If customer wants to use another card for payment then he enters details of that card. System asks for final confirmation to place an order. Customer places an order. System returns confirmation number. A delivery order must exist. The supplier needs to log in, Log-in usecase is carried out. The System displays the delivery order made by customer to supplier. The supplier checks the order. The supplier determines that items are available in inventory to satisfy delivery order. If item is out of stock then the supplier rejects the delivery of that order by asking system to select another supplier. The System reserves the selected items for a customer. The System displays the inventory information to Supplier. The System confirms that the items are reserved to be delivered. To check the status of the order, the customer must have placed an order. Customer enters login id with password, Log-in use case is carried out. The system validates the customer. System presents main page. The system displays a list of available options. The customer selects check order status from the list. The system shows all the available orders. If there is no active order, then the system displays a message that no active order is available. The customer selects the order he wants to check for status. The system displays the status of that order. The customer clicks on OK button. The system shows Thank you message.

Online Pet Store (SNL)

The system shall provide customer with the ability to click the login button on the home page. The system shall be able to display the login page. The system shall provide customer with the ability to enter his login ID with password. The system shall provide customer with the ability to click on the confirm button. The system shall be able to check that all of the required information entered. If the entered information is wrong then the system shall be able to ask customer to reenter details. The system shall be able to validate the entered information against the tables stored in the database. The system shall be able to ensure the customer browser is linked to the supplier catalog website before the starting of the process. The system shall provide customer with the ability to request to browse catalog. The system shall be able to display login page, Login use case is carried out. If the customer is not registered then the system shall be able to ask customer to get registered. The system shall provide customer with the ability to enter email address with a password. The system shall be able to confirm correct login. If the entered email address with password is incorrect then the system shall be able to show login error message. The system shall be able to display catalog information to customer. The system shall be able to display a catalog consisting of the categories of pets: dogs, cats, and birds. The system shall be able to ask user to select the pet category. The system shall provide customer with the ability to select the pet category, browses items. The system shall provide customer with the ability to select the items for pet category from the catalog. If customer does not select any pet category from the displayed catalog then the system shall be able to ask customer to browse related items for pets from the catalog. The system shall be able to display a list of items with price. The

system shall be able to ensure the customer is logged in before placing an order, Log-in use case carried out. The system shall provide customer with the ability to select the items for his pet. The system shall be able to present detailed information about the items. The system shall provide customer with the ability to add the selected item to a shopping cart. The system shall be able to present shopping cart contents to the customer. The system shall provide customer with the ability to select proceed to checkout option. The system shall provide customer with the ability to enter shipping address. The system shall be able to present shipping options. The system shall provide customer with the ability to select shipping option. The system shall be able to ask which credit/debit card used. The system shall provide customer with the ability to select to use the credit card which is stored in the system. If customer wants to use another card for payment then the system shall provide customer with the ability to enter the details of that card. The system shall be able to ask for final confirmation to place an order. The system shall provide customer with the ability to place an order. The system shall be able to return confirmation number. The system shall be able to ensure that a delivery order exists. The system shall be able to ensure the supplier is logged in, Log-in use case carried out. The system shall be able to display the delivery order made by customer to supplier. The system shall provide supplier with the ability to check the order. The system shall provide supplier with the ability to determine that items are available in inventory to satisfy the delivery order. If item is out of stock then the system shall provide supplier with the ability to reject the delivery of that order by asking the system to select another supplier. The system shall be able to reserve the selected items for a customer. The system shall be able to display the inventory information to supplier. The system shall be able to confirm that the items reserved to delivered. The system shall be able to ensure the customer has placed an order. The system shall provide customer with the ability to enter login ID with password, Log-in use case carried out. The system shall be able to validate the customer. The system shall be able to present main page. The system shall be able to display a list of available options. The system shall provide customer with the ability to select check order status from the list. The system shall be able to show all the available orders. If there is no active order, then the system shall be able to display a message that no active order available. The system shall provide customer with the ability to select the order customer wants to check for status. The system shall be able to display the status of that order. The system shall provide customer with the ability to click on OK button. The system shall be able to show thank you message.

Online Pet Store Class Diagram



Online Pet Store Sequence Diagram

